

Unit 13 - Shapes on a Coordinate Plane

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The following B.E.S.T. Standards will be covered in this unit:

MA.912.GR.3.1 Determine the weighted average of two or more points on a line.

- *Clarification 1: Instruction includes using a number line and determining how changing the weights moves the weighted average of points on the number line.*

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties and theorems involving circles, triangles or quadrilaterals.

- *Clarification 1: Instruction includes using the distance or midpoint formulas and knowledge of slope to classify or justify definitions, properties and theorems.*

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles and quadrilaterals.

- *Clarification 1: Problems involving lines include the coordinates of a point on a line segment including the midpoint.*
- *Clarification 2: Problems involving circles include determining points on a given circle and finding tangent lines.*
- *Clarification 3: Problems involving triangles include median and centroid.*
- *Clarification 4: Problems involving quadrilaterals include using parallel and perpendicular slope criteria.*

MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons.

MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.

- *Clarification 1: Instruction includes using the Pythagorean Theorem and completing the square.*
- *Clarification 2: Within the Geometry course, key features are limited to the radius, diameter, and the center.*

Unit 13, Lesson 1: Parallel and Perpendicular Lines



Benchmark

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.



Learning Target

- ☐ I can use coordinate geometry to classify parallel and perpendicular lines.



13.1.1 Warm-Up: Artificial Intelligence and Machine Learning

Artificial intelligence enables a machine to simulate human behavior and decisions. Machine learning is when a machine uses artificial intelligence to learn from past data without a programmer having to re-program the machine.

1. Describe one task you would like a machine to be programmed to do.

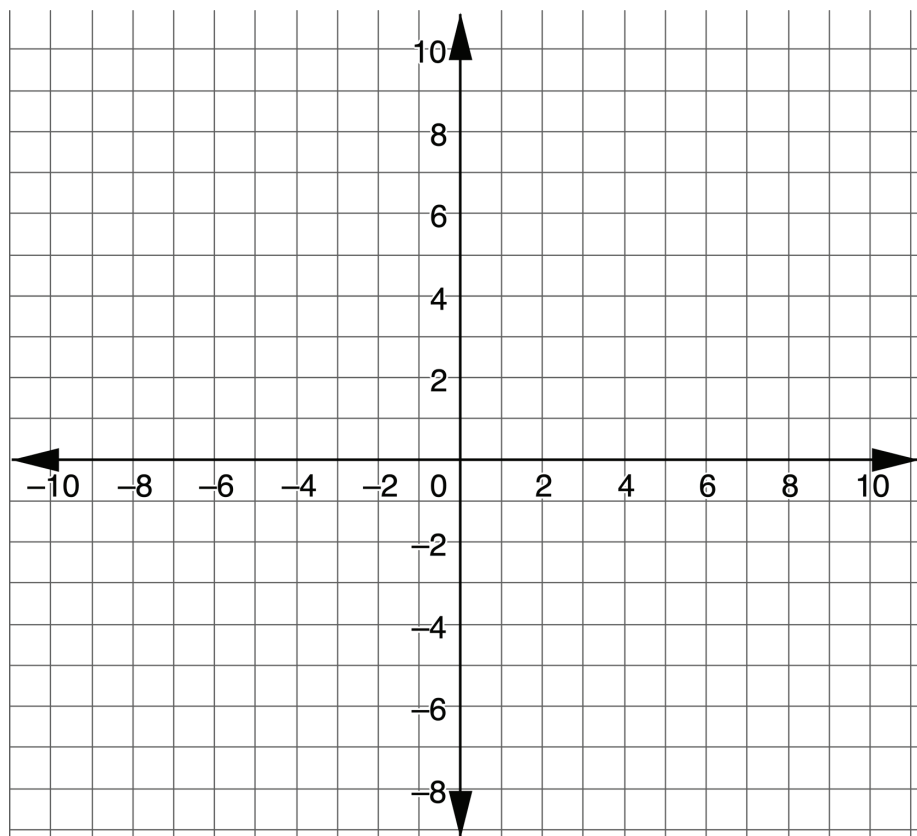
Both artificial intelligence and machine learning rely on mistakes to help improve the programming.

2. Describe a mistake you have made that helped you improve.



13.1.2 Collaboration: Parallel and Perpendicular Lines

1. Graph the equation $y = -2x + 5$ on the coordinate plane.



2. Rewrite the equation $6x + 3y = -3$ in slope-intercept form.

$y =$ _____

3. Graph the equation $6x + 3y = -3$ using slope-intercept form.

4. What similarities do the two equations have?

5. What differences do the two equations have?

6. Complete the statement.

The equations $y = -2x + 5$ and $6x + 3y = -3$ resulted in

- ☐ parallel
- ☐ perpendicular
- ☐ vertical

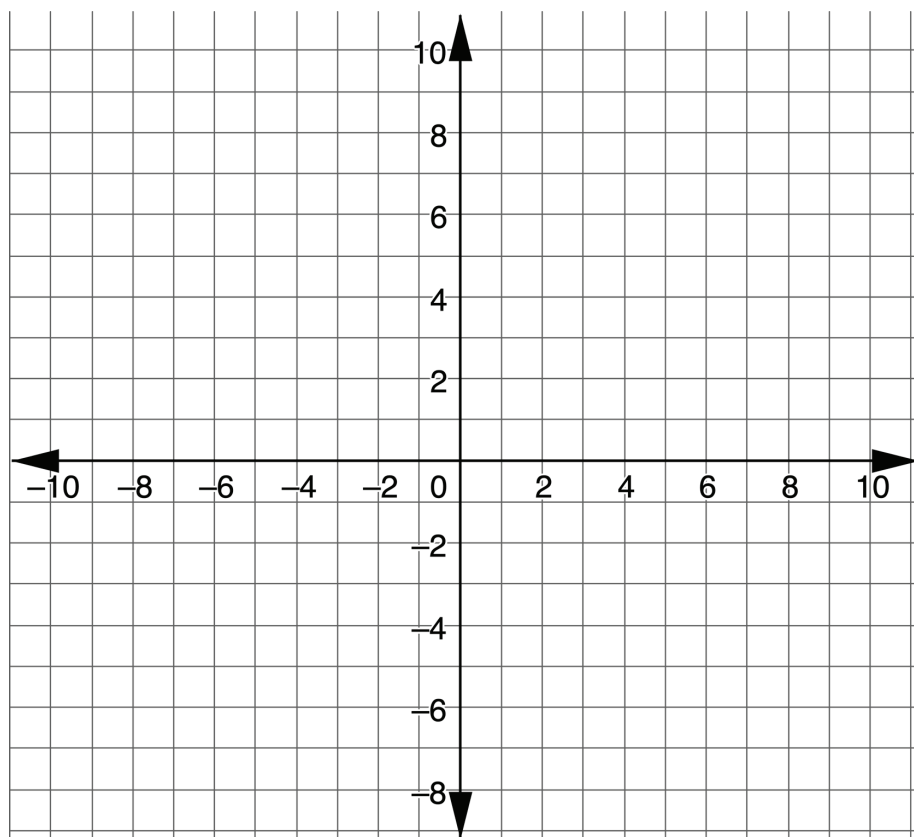
lines. The slopes of the equations are

- ☐ equal.
- ☐ undefined.
- ☐ opposite reciprocals.

7. Rewrite the equation $4y - 6x = 8$ in slope-intercept form.

$y =$ _____

8. Graph the equation $4y - 6x = 8$ using slope-intercept form.



9. Graph the equation $y = -\frac{2}{3}x - 4$.

10. What do you notice about the slopes in the two equations?

11. Complete the statement.

Equation $4y - 6x = 8$ and equation $y = -\frac{2}{3}x - 4$ form

- ☐ parallel
- ☐ perpendicular
- ☐ vertical

lines; therefore, the slopes of the

equations are

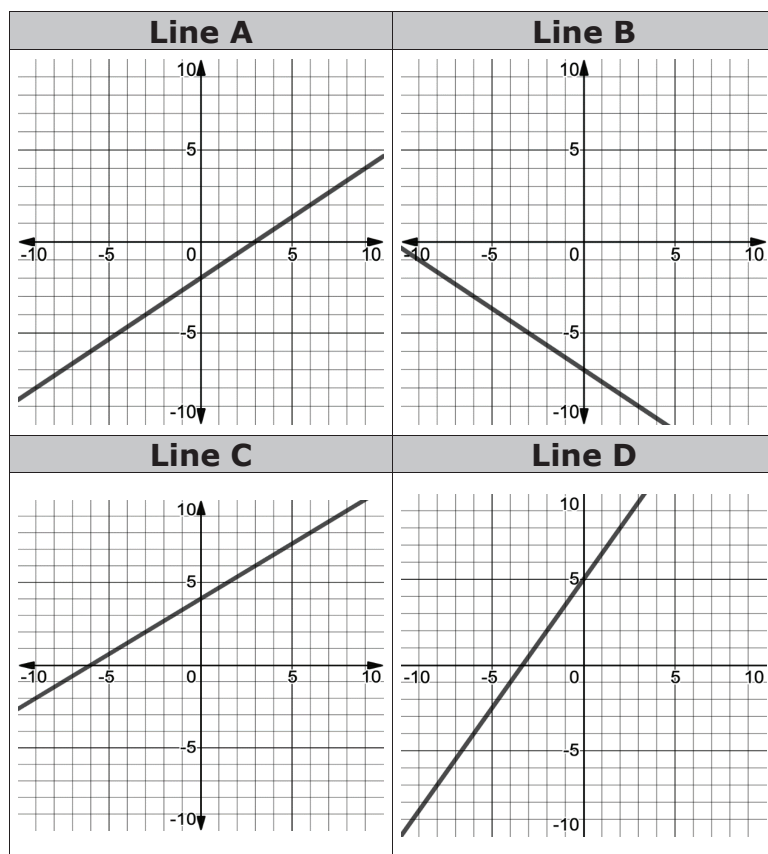
- ☐ equal.
- ☐ undefined.
- ☐ opposite reciprocals.

12. Complete the table.

Type	Sketch	Slope
Parallel lines		
Perpendicular lines		

13. Match the two lines that are parallel and the two lines that are perpendicular. Then, match the equations.

	Lines	Equations
Parallel		
Perpendicular		



Equation E	Equation F
$y + 2 = \frac{2}{3}(x + 9)$	$2x + 3y = -21$
Equation G	Equation H
$-2x + 3y = -6$	$y = \frac{3}{2}x + 5$



13.1.3 Guided Instruction: Parallel and Perpendicular Lines

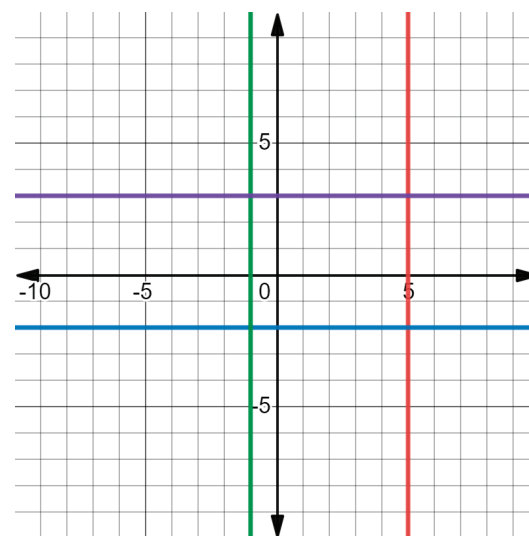


Lines are **parallel** if the slopes of the lines are the same.



Lines are **perpendicular** if the slopes are opposites and reciprocals.

1. Four lines are shown on the coordinate plane.

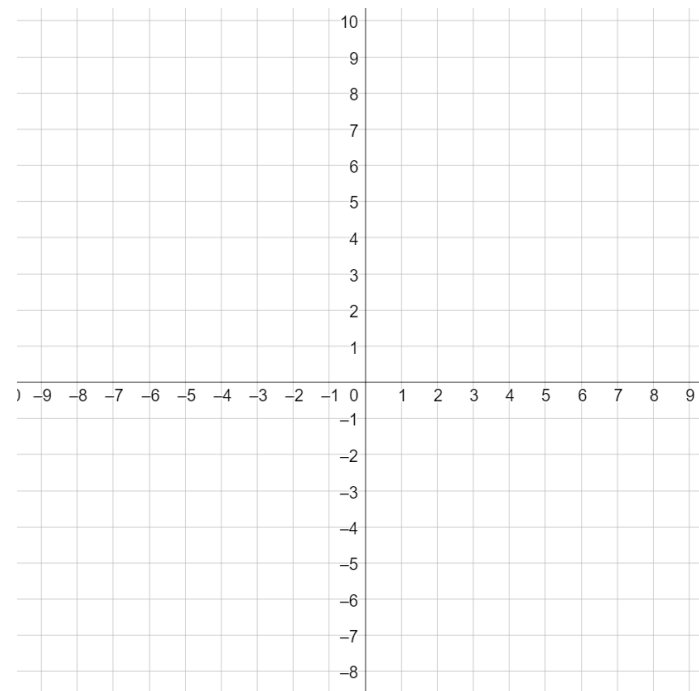


- a. What do you notice about the four lines?

- b. In the graph of the four lines, the four lines create a rectangle. Explain how each of the properties of a rectangle could be shown.

Property	Explanation
Opposite sides are parallel.	
All angles are right angles.	

2. Line W has points $(-1,6)$ and $(1,2)$ on it. Line T has points $(-4,3)$ and $(-2,-2)$ on it.



- a. Graph line W and line T .
- b. Determine if line W and line T are parallel lines, perpendicular lines, or neither. Justify your answer with math.

3. Line N is written as $y = \frac{1}{3}x - 8$. Line V is written as $6x + 2y = 4$.

Determine if line N and line V are parallel lines, perpendicular lines, or neither. Justify your answer with math.



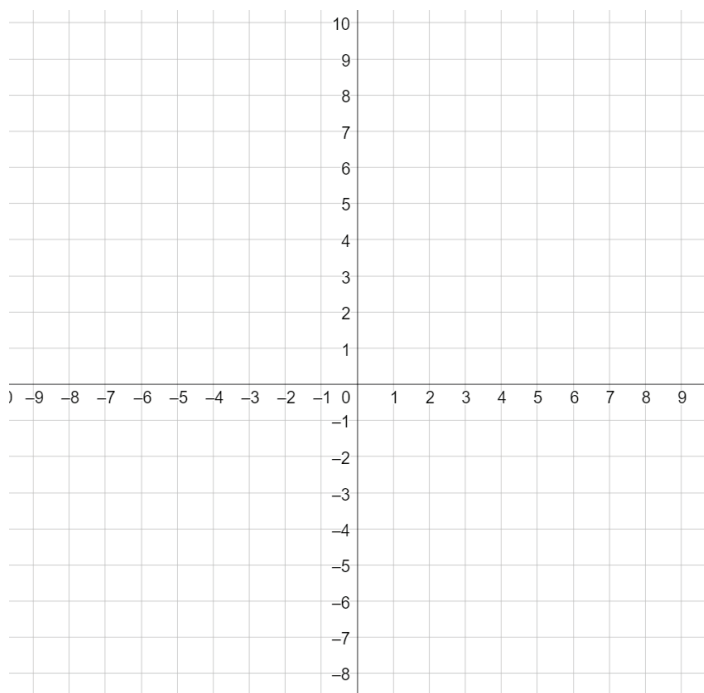
13.1.4 Guided Practice: Parallel and Perpendicular Lines

1. Given that line M is $3x - y = 24$, determine if the following lines are parallel to line M .

Complete the table.

Equation of Line	Work	Is the line parallel to line M ?
$y = 3x + 8$		☐ Yes ☐ No
$y = -3x - 1$		☐ Yes ☐ No
$3y = x - 4$		☐ Yes ☐ No
$3x - y = 3$		☐ Yes ☐ No

2. Line R is written as $4x + y = 5$, line K is written as $12y = 4x - 12$, and line C is written as $4y - 12 = x$.



- Graph lines R , K , and C .
- Determine which two lines are perpendicular lines. Justify your answer with math.



13.1.5 Wrap-Up: Cool Down

- Write a short note to a friend who missed today's class summarizing what you learned.

Unit 13, Lesson 2: Circles on the Coordinate Plane



Benchmarks

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.

MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons.



Learning Targets

- ☐ I can use coordinate geometry to determine parts of a circle in a mathematical context.
- ☐ I can draw a circle using given definitions, properties, or theorems in a mathematical context.
- ☐ I can use coordinate geometry to solve mathematical problems involving perimeter and area of polygons.



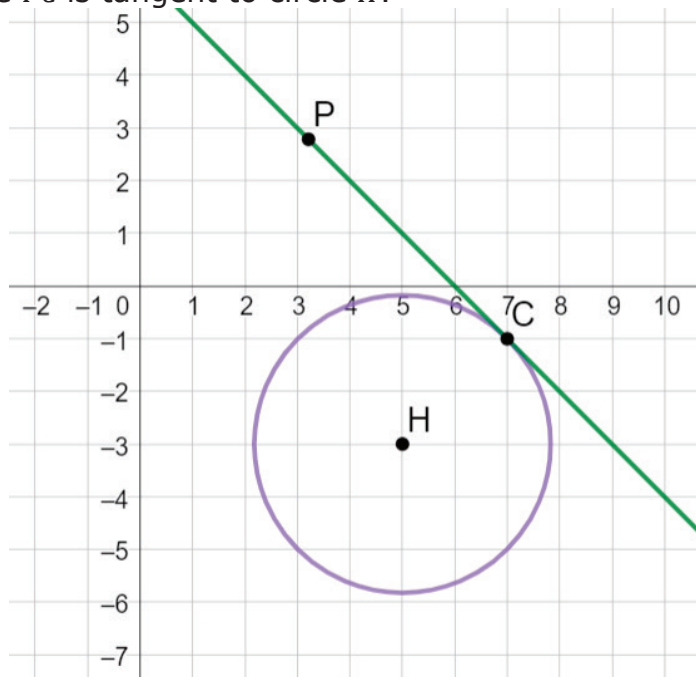
13.2.1 Warm-Up: Bell Work

1. Which equation represents a line that is perpendicular to the line $y = -\frac{1}{3}x + 3$?
 - A. $3x + y = 4$
 - B. $x - 3y = 24$
 - C. $y - 3x = -4$
 - D. $x + 3y = 3$



13.2.2 Exploration: Circles on the Coordinate Plane

1. Circle H is shown, where point C is on the circle and line PC is tangent to circle H .

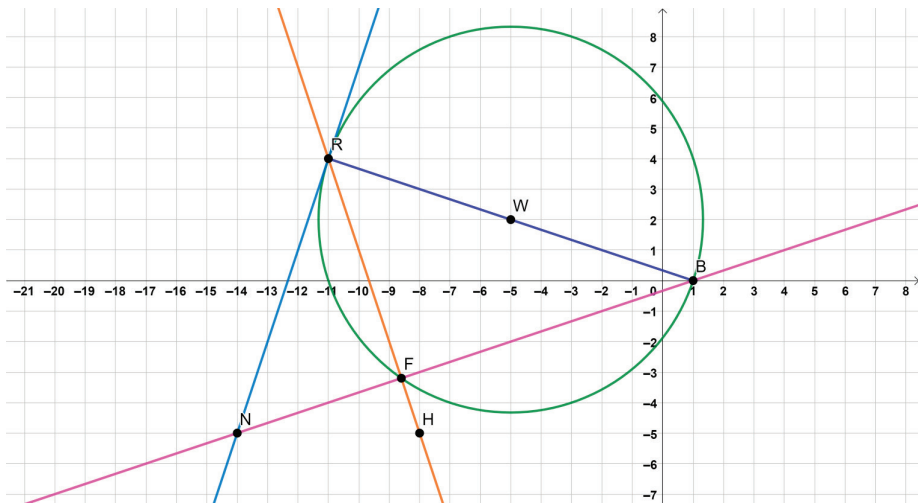


- Use a straightedge to draw a line that goes through points C and H .
- Discuss with your partner how the slope of tangent line PC and the slope of the radius CH are related.
- What property of circles explains this relationship?

- Write the equation of line PC .
- Write the equation of line CH .
- Line PC and line CH intersect at point $C (7, -1)$. Use the equations of lines PC and CH to confirm this point.

2. Circle M has a center at $(-2, 1)$ and a tangent line with the equation $y = 2x - 5$.
- Write the equation of the line perpendicular to the tangent line.
 - Find the point where the tangent line touches circle M . Call this point R .
 - What is the radius of circle M ?
 - Write the equation of circle M .

3. Circle W is shown, with secant lines RH and NB , tangent line NR , and chord RB .



a. Complete the table.

Line Segment	Length
$\overline{NB}$	
$\overline{NR}$	

b. Determine the length of $\overline{NF}$.

c. Complete the table.

Line	Slope	Equation in Point-Slope Form
$\overleftrightarrow{NR}$		
$\overleftrightarrow{RH}$		
$\overleftrightarrow{NB}$		

d. Rewrite the equation of $\overleftrightarrow{RH}$ in slope-intercept form.

e. Rewrite the equation of $\overleftrightarrow{NB}$ in slope-intercept form.

f. Use the slope-intercept forms of $\overleftrightarrow{RH}$ and $\overleftrightarrow{NB}$ to find point F .

- g. Work with your partner to determine if $\triangle BRN$ is isosceles. Show your work as justification.

- h. What could be used to justify that $\triangle RFB$ is a right triangle?

- i. What is the area of $\triangle RFB$?

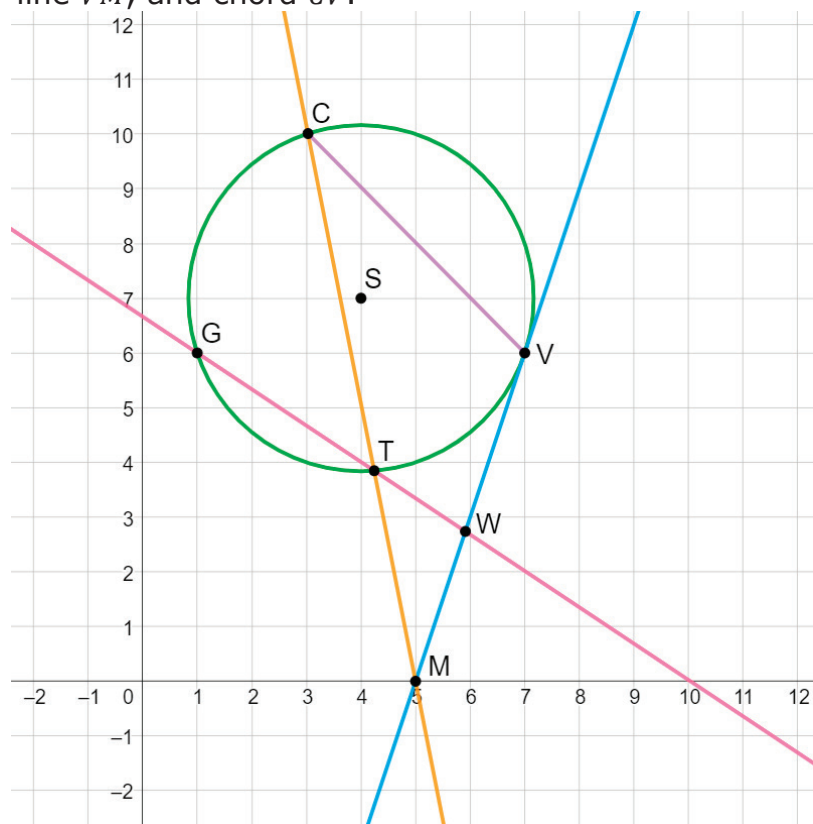
- j. What is the perimeter of $\triangle RFN$?



13.2.3 Try It: Circles on the Coordinate Plane

1. Circle G has a center at $(9, -2)$ and a tangent line with the equation $3x + 2y = 36$.
 - a. At what point does the tangent line touch circle G ?
 - b. Write the equation of circle G .

2. Circle S is shown with secant lines CM and GW , tangent line VM , and chord CV .



- a. Complete the table.

Line Segment	Length
$\overline{VM}$	
$\overline{CV}$	
$\overline{CM}$	
$\overline{MT}$	

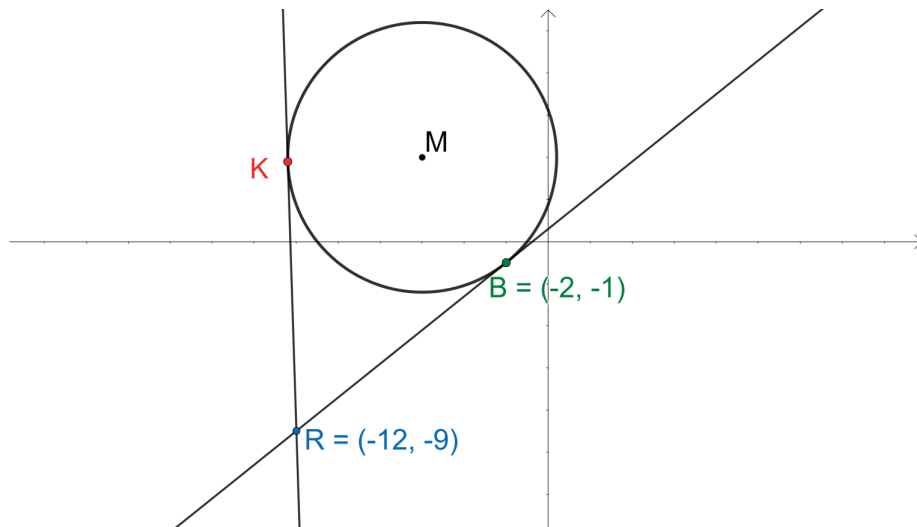
- b. What is the perimeter of $\triangle CMV$?

- c. Complete the table.

Line	$\overleftrightarrow{GW}$	$\overleftrightarrow{CM}$
Slope		
Equation in Point-Slope Form		
Equation in Slope-Intercept Form		

- d. Use the slope-intercept form of both lines to find point T .

3. Circle M is shown with tangents $\overline{RK}$ and $\overline{RB}$.



- What is the length of $\overline{RK}$?
- The center of the circle is located at $(-6, 4)$. Write the equation of circle M .
- What type of quadrilateral is $KMBR$? Justify your choice.
- What is the distance between point R and point M ?



13.2.4 Wrap-Up: Cool Down

- Write a note to a classmate who might have missed today's lesson, in your own words, about circles on the coordinate plane.

Unit 13, Lesson 3: Determining Points on a Circle



Benchmarks

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.

MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.



Learning Targets

- ☐ I can determine whether a point is on a circle, inside a circle, or outside a circle.
- ☐ I can find the center of a circle when given two points.
- ☐ I can write the equation of a circle given the endpoints of a diameter.



13.3.1 Warm-Up: Notice and Wonder

Castlerigg Stone Circle is one of the earliest stone circles to be found in Britain and is important in terms of megalithic astronomy and geometry. It is composed of 38 free standing stones, some measuring up to 10 feet high and dating back to the Neolithic period, 4000 to 5000 years ago.



1. What do you notice?

2. What do you wonder?



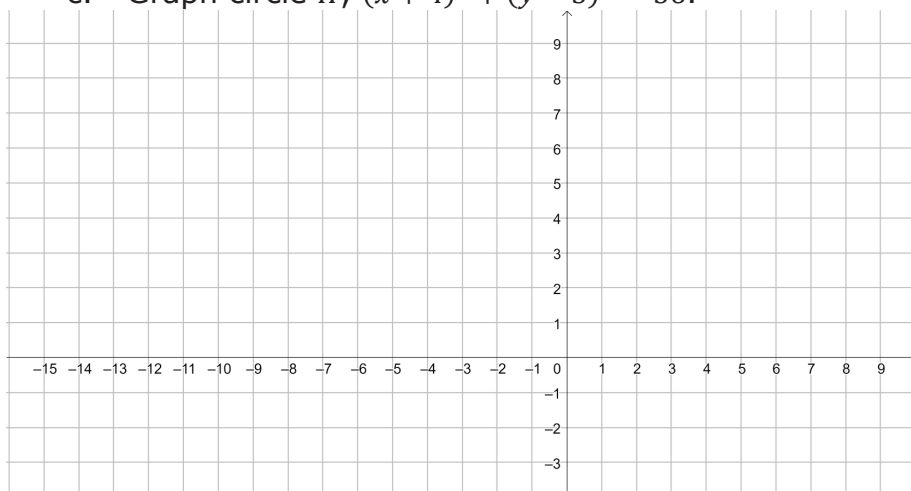
13.3.2 Exploration: Points on a Circle

1. The equation of circle H is $(x + 4)^2 + (y - 3)^2 = 36$.
- a. Complete the table by substituting the given coordinates for x and y in the equation of circle H . Then, determine the value of the left side of the equation to complete the simplified equation in the last column.

Coordinate	Substitution	Simplified Equation
$M(-8, -2)$		$\underline{\hspace{2cm}} = 36$
$B(-3, -1)$		$\underline{\hspace{2cm}} = 36$
$R(-4, -3)$		$\underline{\hspace{2cm}} = 36$

- b. What do you notice about the simplified equations?

- c. Graph circle H , $(x + 4)^2 + (y - 3)^2 = 36$.



- d. Plot points $M(-8, -2)$, $B(-3, -1)$, and $R(-4, -3)$.
- e. What do you notice about the relationship between the location of the points and their simplified equations?
- f. Write a conjecture about the location of a point and the equation of the circle.

- g. Ask two classmates for their conjectures. Record their conjectures and write your summary of their conjecture. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Conjecture	Initials

- h. Review your conjecture with your partner. If necessary, make any changes.

2. The equation of circle K is $(x + 6)^2 + (y + 5)^2 = 25$ with points $N(-6,0)$, $V(-3,-1)$, $T(-1,-5)$, and $R(-9,-9)$ on the circle.

- a. What is the length of the diameter of circle K ?

- b. Complete the table by finding the distance between the given points.

Points	Work	Distance
$N(-6,0)$ and $R(-9,-9)$		
$V(-3,-1)$ and $R(-9,-9)$		
$T(-1,-5)$ and $R(-9,-9)$		

- c. What do you notice about the distances between the points?

- d. What can you conclude about points V and R ?

3. Circle W has a radius of $\sqrt{29}$ and points $S(2, -2)$, $D(5, 1)$, and $H(12, -6)$.

a. Determine which two points are the endpoints of a diameter of circle W . Show your work.

b. Describe how to find the center of the circle using the two endpoints of the diameter.

c. What is the center of the circle?

d. Write the equation for circle W .

4. The points $G(3, -1)$ and $H(3, 3)$ are the endpoints of a diameter of circle M . Write the equation of circle M .





13.3.3 Try It: Points on a Circle

1. The equation of circle P is $(x - 8)^2 + (y - 2)^2 = 13$.

Determine whether the following points lie inside, outside, or on circle P .

Point	Work	Location
$(5, -1)$		☐ Inside circle P ☐ Outside circle P ☐ On circle P
$(6, 5)$		☐ Inside circle P ☐ Outside circle P ☐ On circle P
$(12, 2)$		☐ Inside circle P ☐ Outside circle P ☐ On circle P
$(11, 0)$		☐ Inside circle P ☐ Outside circle P ☐ On circle P
$(9, 4)$		☐ Inside circle P ☐ Outside circle P ☐ On circle P

2. Circle T and circle R intersect at point H .

Equation of circle T is $x^2 + (y - 7)^2 = 20$

Equation of circle R is $(x - 7)^2 + (y - 3)^2 = 25$

- a. Which coordinate pair is the location of point H ?
- ☐ $(-4, 5)$
 - ☐ $(7, 8)$
 - ☐ $(2, 3)$
 - ☐ $(4, 9)$

3. Circle S has a radius of 10 and points $F(-6, -6)$, $C(8, 8)$, $W(10, 2)$, and $M(-8, -4)$ on the circle.

- a. Determine which two of the given points are endpoints of a diameter of circle S .
- b. Find the center of circle S .
- c. Write the equation of circle S .



13.3.4 Wrap-Up: Ticket Out the Door

1. The equation of circle W is $x^2 + (y + 4)^2 = 26$.

Select all the points that lie on circle W .

- ☐ (3,0)
- ☐ (5, -3)
- ☐ (1,1)
- ☐ (-3, -2)
- ☐ (-5, -5)

Unit 13, Lesson 4: Partitioning Line Segments



Benchmark

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.



Learning Target

- ☐ I can use coordinate geometry to partition a line segment.



13.4.1 Warm-Up: Heroic Person

1. Who do you consider heroic and inspirational?
2. Why did you pick this person?



13.4.2 Exploration: Partitioning a Line Segment

1. Directed line segment $\overline{SP}$ is shown.



- a. How many equal parts is $\overline{SP}$ partitioned into?

- b. Make a mark where $\overline{SP}$ is partitioned into a 2:3 ratio. Call this point R .

- c. Complete the statement.

Point R is

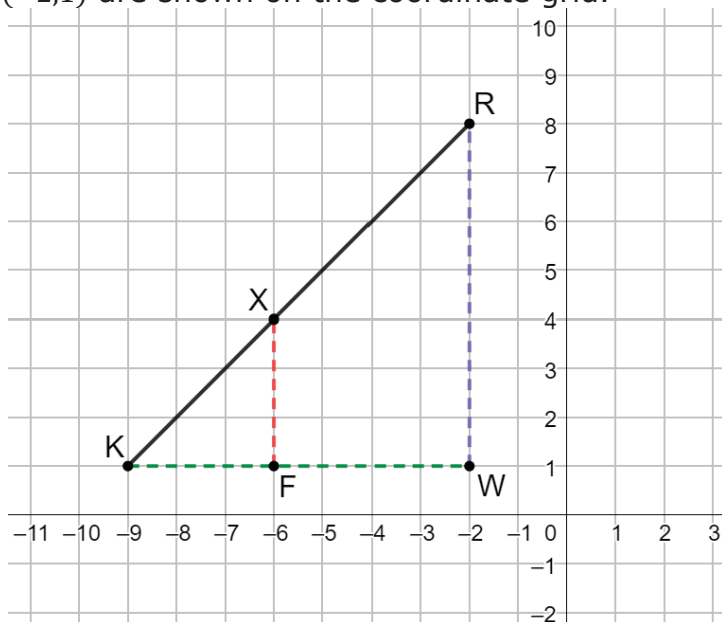
☐ $\frac{2}{3}$
☐ $\frac{2}{5}$
☐ $\frac{3}{5}$

of the distance from S to P .

- d. Complete the equation that can be used to find the distance of $\overline{SR}$.

$$SR = \underline{\hspace{1cm}} SP$$

2. Point X partitions directed line segment KR in a 3:4 ratio. Points $K(-9,1)$, $F(-6,1)$, $X(-6,4)$, $R(-2,8)$, and $W(-2,1)$ are shown on the coordinate grid.



- a. Use the distances to write the numeric ratio for each segment ratio. Then, choose the type of ratio the segment ratio represents.

Segment Ratio	Numeric Ratio	Type of Ratio
$KF:FW$		☐ part:part ☐ part:whole
$KF:KW$		☐ part:part ☐ part:whole
$XF:RW$		☐ part:part ☐ part:whole
$KX:KR$		☐ part:part ☐ part:whole

- b. What do you notice about the numeric ratios?

- c. Two students were discussing the directed line segment KR .

Nevaeh	Isabella
Point R is a dilation of point X centered at point K using a scale factor of $\frac{3}{4}$.	Point R is a dilation of point X centered at point K using a scale factor of $\frac{3}{7}$.

Work with your partner to explain why neither student has the correct scale factor. Include in your explanation what the correct scale factor should be so that the result of dilating at point K is point R .

- d. Discuss with your partner which ratios can be used to justify that $\triangle KXF \sim \triangle KRW$. Summarize your discussion.
- e. With your partner, write a conjecture on how point X could be found if only given the endpoints of the directed line segment KR and the ratio.

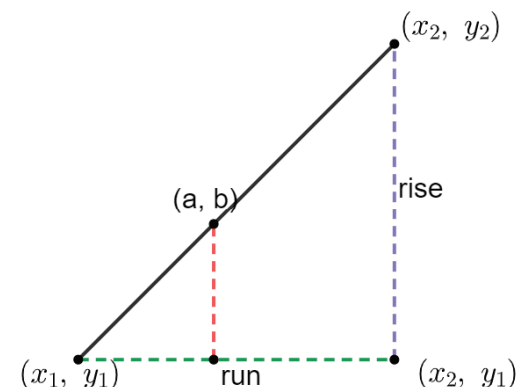
- f. Ask two other pairs of partners for the conjecture they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

Conjecture	Initials

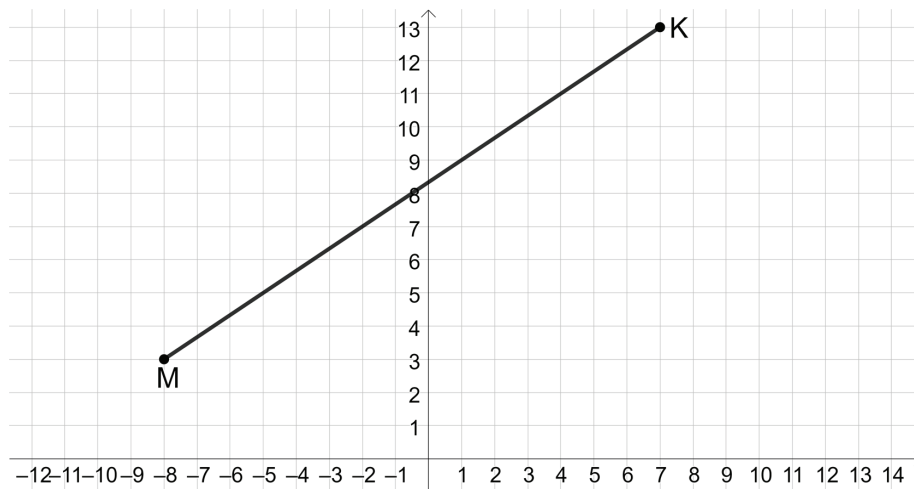


13.4.3 Guided Instruction: Partitioning a Line Segment

1. Generalize the conjecture in the Exploration if the given ratio is part:whole.



2. Directed line segment MK is shown with endpoints at point $M(-8,3)$ and point $K(7,13)$.



Directed line segment MK is partitioned by point B in a ratio of 2:3.

- Write an equation that can be used to find the distance of $\overline{MB}$.
- Use your equation to find the horizontal distance of $\overline{MB}$.

- Use your equation to find the vertical distance of $\overline{MB}$.
- Use the horizontal distance and the vertical distance to plot point B on $\overline{MK}$.
- What are the coordinates of point B ?

3. Directed line segment XY has endpoints at point $X(-6,1)$ and point $Y(4,4)$.



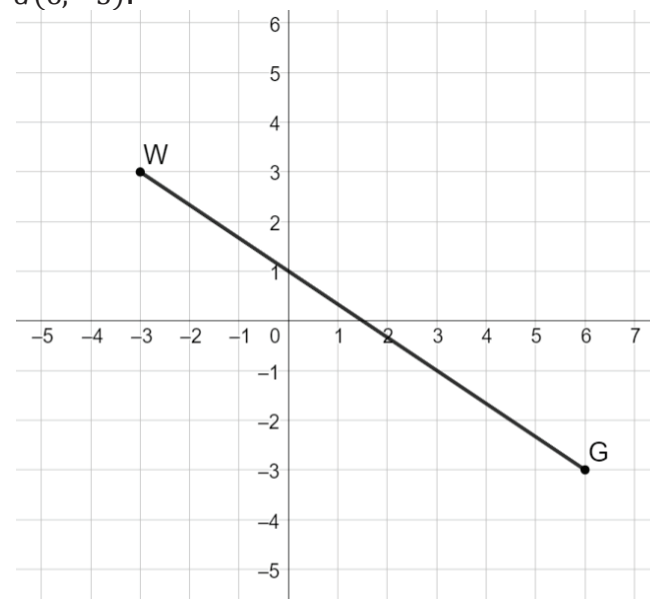
Directed line segment XY is partitioned $\frac{1}{2}$ of the distance from X to Y at point W .

- What is the horizontal distance of $\overline{XW}$?
- What is the vertical distance of $\overline{XW}$?
- What are the coordinates of point W ?



13.4.4 Try It: Partitioning a Line Segment

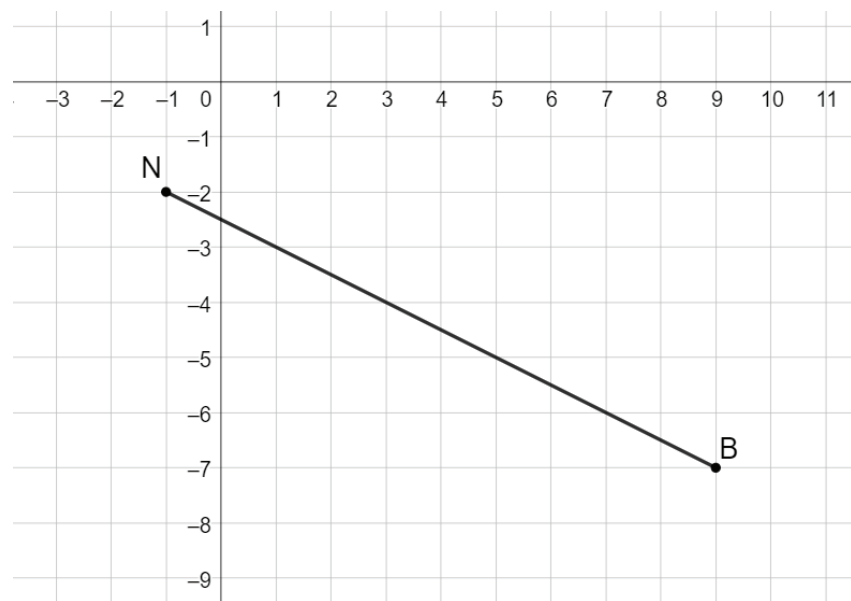
1. Directed line segment WG has endpoints at $W(-3,3)$ and $G(6,-3)$.



Directed line segment WG is partitioned $\frac{2}{3}$ of the distance from W to G at point D .

- What is the horizontal distance of $\overline{WD}$?
- What is the vertical distance of $\overline{WD}$?
- What are the coordinates of point D ?

2. Directed line segment BN has endpoints at point $B(9, -7)$ and point $N(-1, -2)$.



Directed line segment BN is partitioned by point H at a 4:1 ratio.

What are the coordinates of point H ?



13.4.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Unit 13, Lesson 5: Weighted Averages



Benchmark

MA.912.GR.3.1 Determine the weighted average of two or more points on a line.



Learning Targets

- ☐ I can determine the weighted average of two or more points on a line.
- ☐ I can explain how changing the weight moves the weighted average of points.



13.5.1 Warm-Up: Between Two Numbers

A stock exchange, or stock market, is a system for buying and selling bonds, securities, or stocks. A stock is a share in the ownership of a company. NASDAQ is short for National Association of Securities Dealers Automated Quotations, which is the world's largest electronic stock exchange. Each company that is traded has an abbreviation of one to four letters.

Jaxon's grandmother bought 100 shares of Netflix and 100 shares of Apple for him when he was born on March 29, 2003. She paid \$1.37 per share for Netflix (NASDAQ: NFLX) and paid \$0.21 per share for Apple (NASDAQ: AAPL).

On Jaxon's 18th birthday, Netflix's stock price was at \$513.95 and Apple's stock price was at \$121.39.

1. How much were the stocks worth on Jaxon's 18th birthday?

The stock price history of Netflix shows that at one point the stock had increased in worth by 45,000%, and Apple stock had increased in worth by 140,000%.

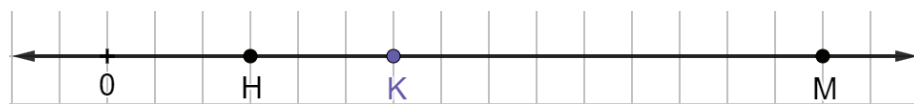
2. Write the decimal equivalent of each percentage increase.

Stock	Decimal Equivalent
Apple	
Netflix	



13.5.2 Exploration: Exploring Weighted Averages

1. A number line is shown. Each grid space is 1 unit.



- a. Find the midpoint of $\overline{HM}$ and label it B .

Assuming that points H and M have the same weight, the diagram represents how points H , B , and M are in balance.



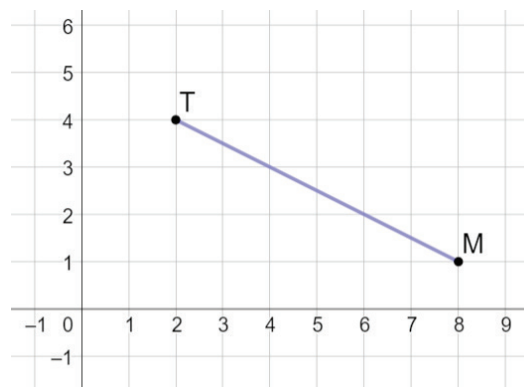
- b. Discuss with your partner why point B acts as a point of balance. Summarize your discussion.
- c. Keyshawn told his partner that if the scale is drawn where the scale is centered on point K , then point H would be higher than point M . Describe or draw what would have to be done to balance the scale when it is centered at point K .

The ratio of $HK:KM$ is 3:9, which can be written as $\frac{1}{3}$. The ratio, $\frac{1}{3}$, can be used to determine the location of point K using the steps shown.

$$\frac{3}{4}(\text{point } H) + \frac{1}{4}(\text{point } M) = \frac{3}{4}(3) + \frac{1}{4}(15) = \frac{9}{4} + \frac{15}{4} = \frac{24}{4} = 6$$

The ratio $\frac{1}{3}$ was used to find $\frac{3}{4}$ and $\frac{1}{4}$ by noting that the ratio indicates there are a total of 4 parts. To balance at point K , point H would be "heavier" than point M , so it is weighted as 3 of the 4 parts with point M weighted as 1 of the 4 parts.

2. Point T has coordinates $(2, 4)$, and point M has coordinates $(8, 1)$.



- a. What is the midpoint of line segment TM ?

b. Complete the table.

	<i>x</i> -value	<i>y</i> -value
<i>T</i> (2, 4)	$\frac{1}{2}(\rule{1cm}{0.4pt})$	$\frac{1}{2}(\rule{1cm}{0.4pt})$
<i>M</i> (8, 1)	$\frac{1}{2}(\rule{1cm}{0.4pt})$	$\frac{1}{2}(\rule{1cm}{0.4pt})$
Total		

- c. Discuss with your partner how the table shows the process of finding the midpoint.
- d. Complete the expression that can be used to find the point that partitions the line segment in a 1:1 ratio. *T* represents the coordinates of point *T*, and *M* represents the coordinates of point *M*.

$$\frac{1}{\rule{1cm}{0.4pt}}T + \frac{1}{\rule{1cm}{0.4pt}}M$$

- e. Find the point that partitions directed line segment *TM* in a 2:1 ratio.

f. Complete the table to calculate $\frac{1}{3}T + \frac{2}{3}M$.

	<i>x</i> -value	<i>y</i> -value
<i>T</i> (2, 4)	$\frac{1}{3}(\rule{1cm}{0.4pt})$	$\frac{1}{3}(\rule{1cm}{0.4pt})$
<i>M</i> (8, 1)	$\frac{2}{3}(\rule{1cm}{0.4pt})$	$\frac{2}{3}(\rule{1cm}{0.4pt})$
Total		

- g. What do you notice about the total in the table and the point found in part e that partitions the directed line segment in a 2:1 ratio?

3. Line segment *KR* has endpoints at *K*(−6, 1) and *R*(2, 5).

- a. Complete the expression for the point that partitions directed line segment *KR* in a 3:1 ratio.

$$\frac{\rule{1cm}{0.4pt}}{4}K + \frac{\rule{1cm}{0.4pt}}{4}R$$

- b. Use the expression to find the point that partitions directed line segment *KR* in a 3:1 ratio.



13.5.3 Guided Instruction: Determining Weighted Averages

Weight is a measure of force pushing or pulling on an object due to gravity. This means that finding a weighted average puts more “weight” on one point than another.

If B is weighted twice as heavily as A , then during the calculation, B is to measure two times the value of A . In other words, B has two times the “gravitational pull” of A .

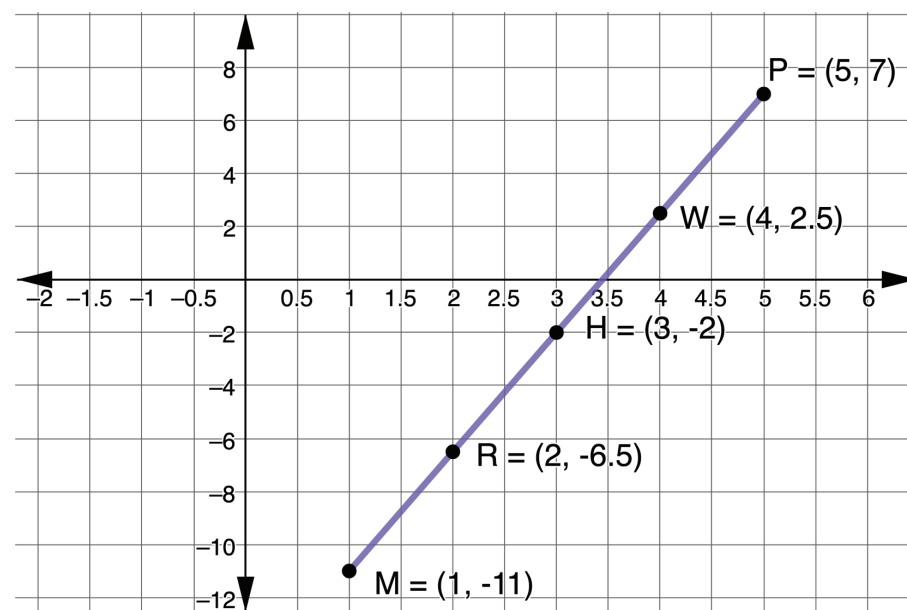
Think of it in terms of a seesaw. In the picture below, point B is weighted more heavily than point A .



1. Discuss with your partner what the girls close to point B would have to do so that the seesaw would balance. Summarize your discussion.



Directed line segment $\overline{MP}$ is shown, where point M is at $(1, -11)$ and point P is at $(5, 7)$. Point H is the midpoint of $\overline{MP}$, point W is the midpoint of $\overline{HP}$, and point R is the midpoint of $\overline{MH}$.



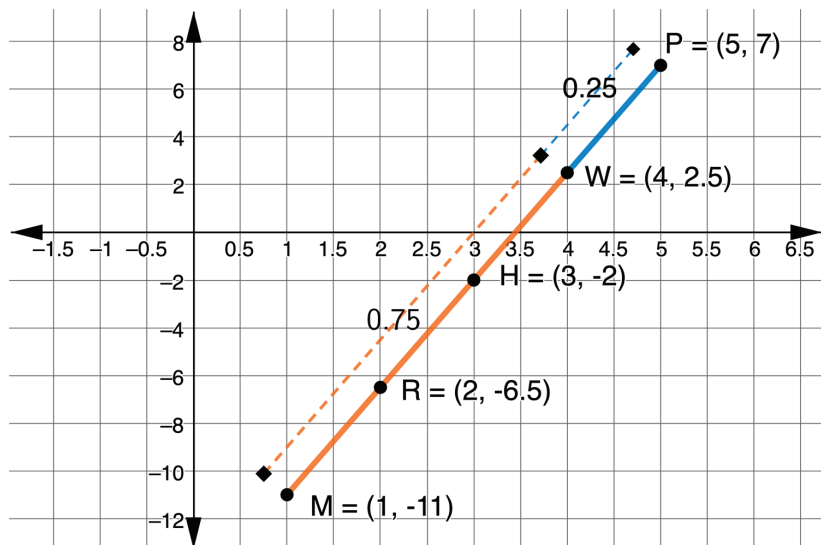
Since point H is the midpoint of $\overline{MP}$, the coordinates of point H can be determined by finding the average of points M and P .

$$\frac{M+P}{2}$$

Point W partitions the line segment $\overline{MP}$ so that point P weighs three times as much as point M . This means that point W is closer to point P because point P is “heavier” than point M . The given expression can be used to calculate this weighted average when point P weighs three times as much as point M .

$$\frac{M+P+P+P}{4}$$

The graph shows the weighted averages.



2. Discuss with your partner why the distance from point W to point P is shorter if point P is "heavier" than point M . Summarize your discussion.

3. Point R partitions the line segment MP so that point M weighs three times as much as point P . Write an expression that shows this weighted average.



A midpoint calculation weighs points on a line segment equally. In a **weighted average** calculation, points on a line segment can be weighted differently.

4. Write the expression that can be used to find the point that partitions $\overline{MP}$ in the given ratio, and find the endpoint that is the weighted point.

Ratio	Expression	Point	Weighted Point
1:1			
1:3			
3:1			

5. A line segment is shown in the table with three different weighted averages. Write the expression that can be used to find each weighted average.

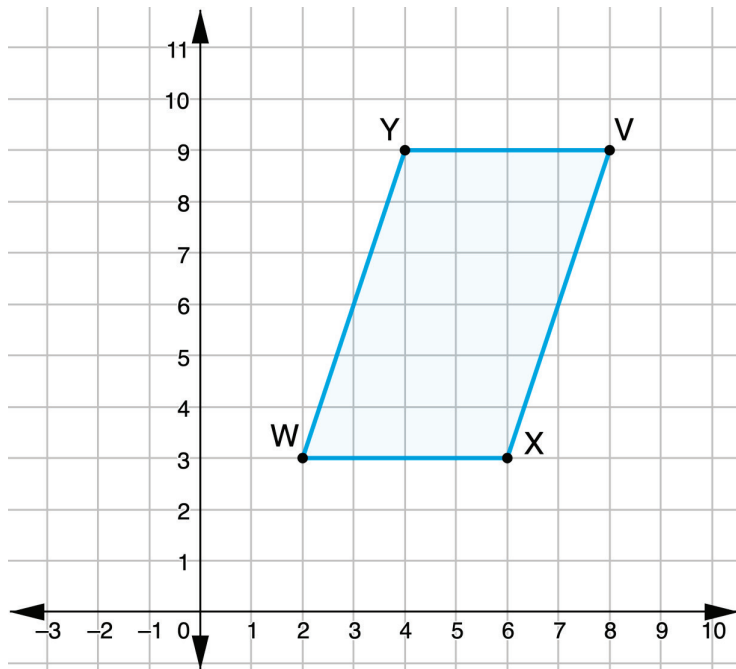
Partition	Graph	Expression
4:1		
2:3		
1:5		

6. Discuss with your partner how changing the weights moves the weighted average on the number line. Summarize your discussion.



13.5.4 Collaboration: Weighted Averages and Dilations

1. $VXWY$ has vertices at $V(8,9)$, $X(6,3)$, $W(2,3)$, and $Y(4,9)$.

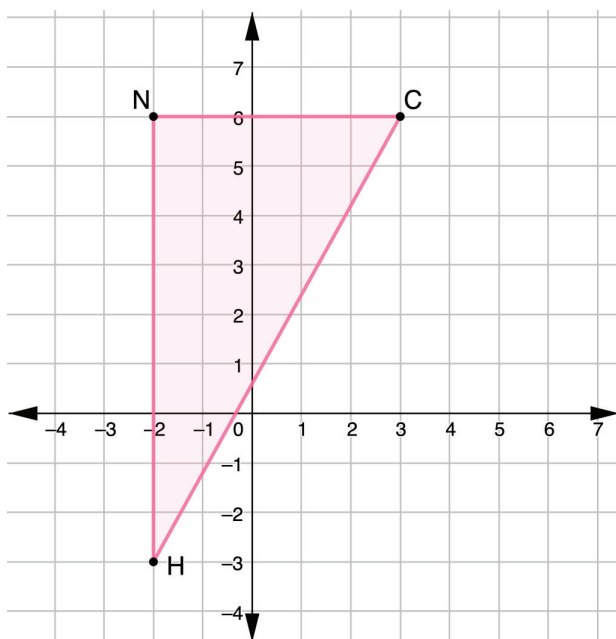


- Find the point that partitions $\overline{WX}$ in a 1:4 ratio. Label it R .
- Find the point that partitions $\overline{WY}$ in a 1:4 ratio. Label it T .

- Find the point that partitions $\overline{VV}$ in a 1:4 ratio. Label it S .

- Is $WTSR$ a dilation of $WYVX$? Justify your answer.

2. Triangle NCH has vertices at $N(-2,6)$, $C(3,6)$, and $H(-2,-3)$.



Use segment partitioning to dilate $\triangle NCH$ using center C and scale factor $\frac{3}{4}$.



13.5.5 Your Turn!

1. Line segment MK has endpoints at $M(-2, 1)$ and $K(8, 7)$.
 - a. Write an expression for the point that partitions directed line segment MK in a 2:3 ratio.
 - b. Use the expression to find the point that partitions directed line segment MK in a 2:3 ratio.



13.5.6 Wrap-Up: Ticket Out the Door

1. A number line is shown. Each grid space is 1 unit.



- a. Determine the point on the number line that satisfies the expression shown. Label the point M .

$$\frac{2}{5}(\text{point } W) + \frac{3}{5}(\text{point } T)$$

- b. What is the ratio $WM:MT$?

Unit 13, Lesson 6: Classifying Triangles



Benchmark

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.



Learning Targets

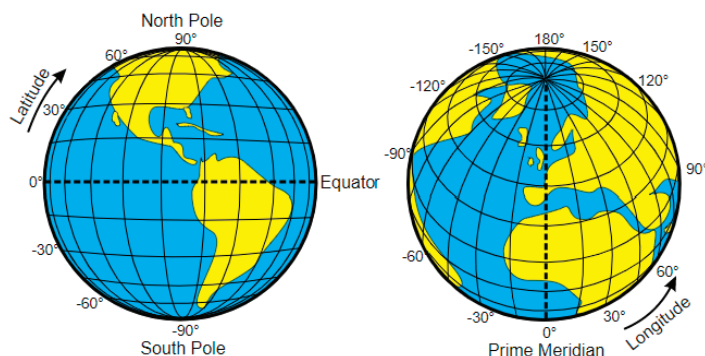
- ☐ I can use coordinate geometry to classify triangles in a mathematical or real-world context.
- ☐ I can draw a triangle on a coordinate plane using given definitions, properties, or theorems.



13.6.1 Warm-Up: Latitude and Longitude

In ancient times, people used landmarks to identify locations, but travelers were not familiar with local landmarks, making traveling difficult. Hipparchus, a Greek astronomer, was the first to specify location using latitude and longitude as coordinates. Hipparchus applied his knowledge of spherical angles to the problem of denoting locations on the Earth's surface.

Latitude is the measurement of the globe north to south, and longitude is the measurement of the globe east to west.



The table shows the latitude and longitude of different Florida cities.

City	Coordinates	City	Coordinates
Panama City	30.2N, 85.7W	Miami	25.8N, 80.2W
Tallahassee	30.46N, 84.28W	Tampa	28.0N, 82.5W
Lake Worth	26.6N, 80.1W	Orlando	28.5N, 81.4W
Jacksonville	30.3N, 81.7W	Gainesville	29.7N, 82.3W

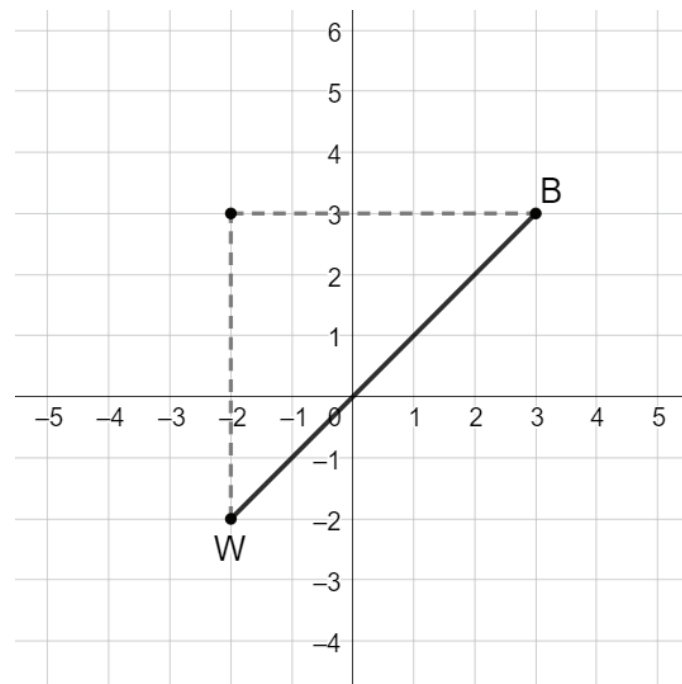
1. What do you notice? What do you wonder?



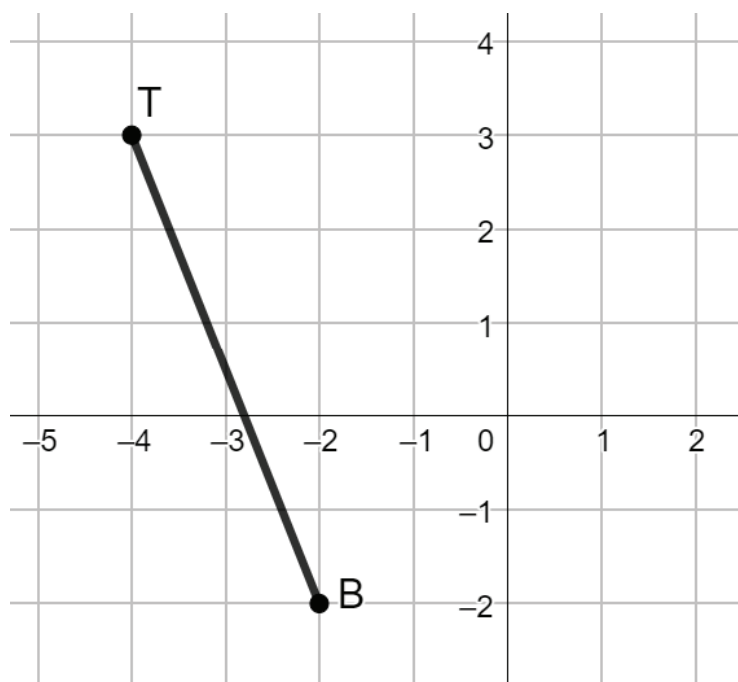
13.6.2 Refresher: Finding Distance

Work with your partner to complete the following.

1. Find the length of $\overline{WB}$ using the Pythagorean Theorem. If necessary, round to the nearest thousandth.



2. Use the Pythagorean Theorem to find the length of $\overline{TB}$. If necessary, round to the nearest thousandth.



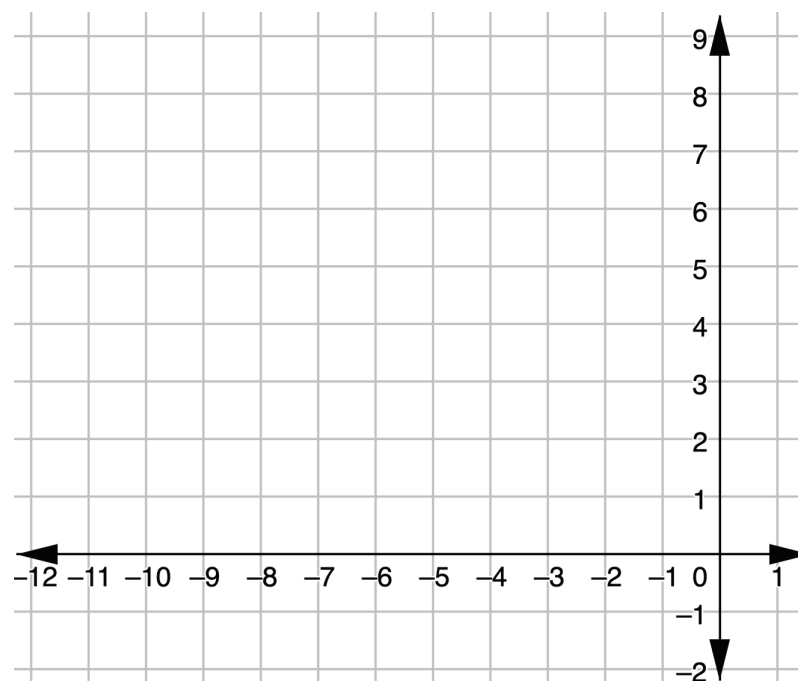
3. Explain how the Pythagorean Theorem can be used to find the distance between two points.



13.6.3 Exploration: Classifying Triangles

1. $\triangle MKC$ has vertices at $M(-8,8)$, $K(-4,2)$, and $C(-9,3)$.

- a. Graph $\triangle MKC$ on the coordinate plane.



- b. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Length of Line Segment
$\overline{CM}$	
$\overline{MK}$	
$\overline{KC}$	

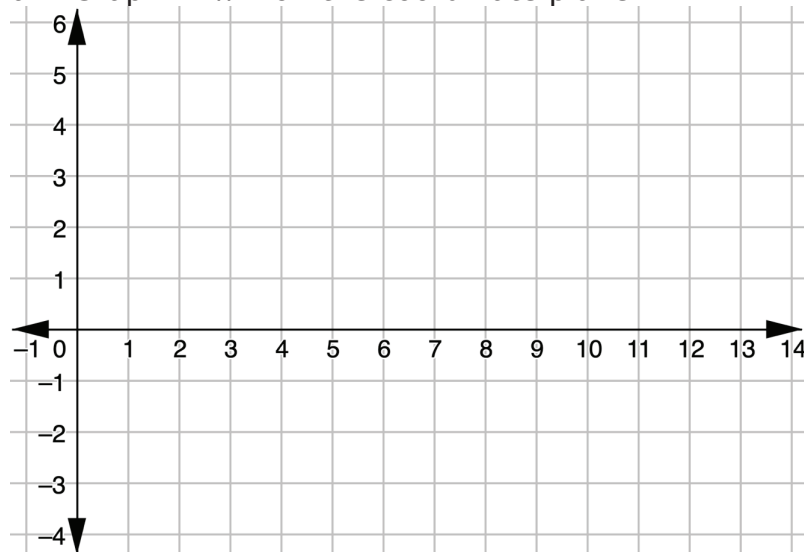
c. What do you notice about the side lengths of $\triangle MKC$?

d. Complete the statement.

$\triangle MKC$ is classified as a(n)
☐ equilateral
☐ isosceles
☐ scalene
 triangle
 because $\triangle MKC$ has
☐ no
☐ two
☐ three
 equal side lengths.

2. $\triangle NWR$ has vertices at $N(6,5)$, $W(3,-1)$, and $R(10,-1)$.

a. Graph $\triangle NWR$ on the coordinate plane.



b. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Length of Line Segment
$\overline{NR}$	
$\overline{RW}$	
$\overline{WN}$	

c. What do you notice about the side lengths of $\triangle NWR$?

d. Complete the statement.

$\triangle NWR$ is classified as a(n)
☐ equilateral
☐ isosceles
☐ scalene
 triangle
 because $\triangle NWR$ has
☐ no
☐ two
☐ three
 equal side lengths.

3. $\triangle MSG$ has vertices at $M(-3, 4)$, $S(2.2, 1)$, and $G(-3, -2)$.

- a. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Length of Line Segment
$\overline{MS}$	
$\overline{SG}$	
$\overline{GM}$	

- b. Complete the statement.

$\triangle MSG$ is classified as a(n)

- ☐ equilateral
- ☐ isosceles
- ☐ scalene

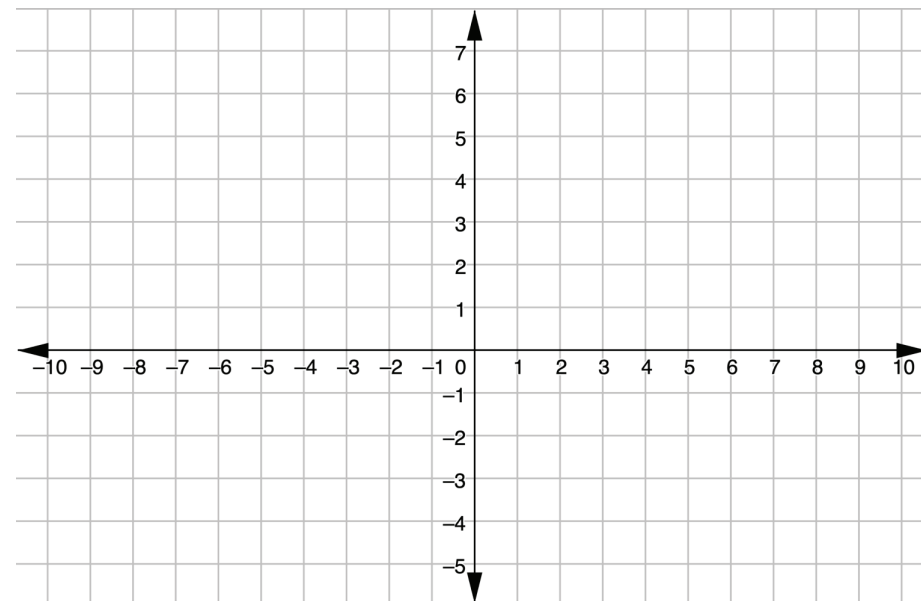
triangle

because $\triangle MSG$ has

- ☐ no
- ☐ two
- ☐ three

equal side lengths.

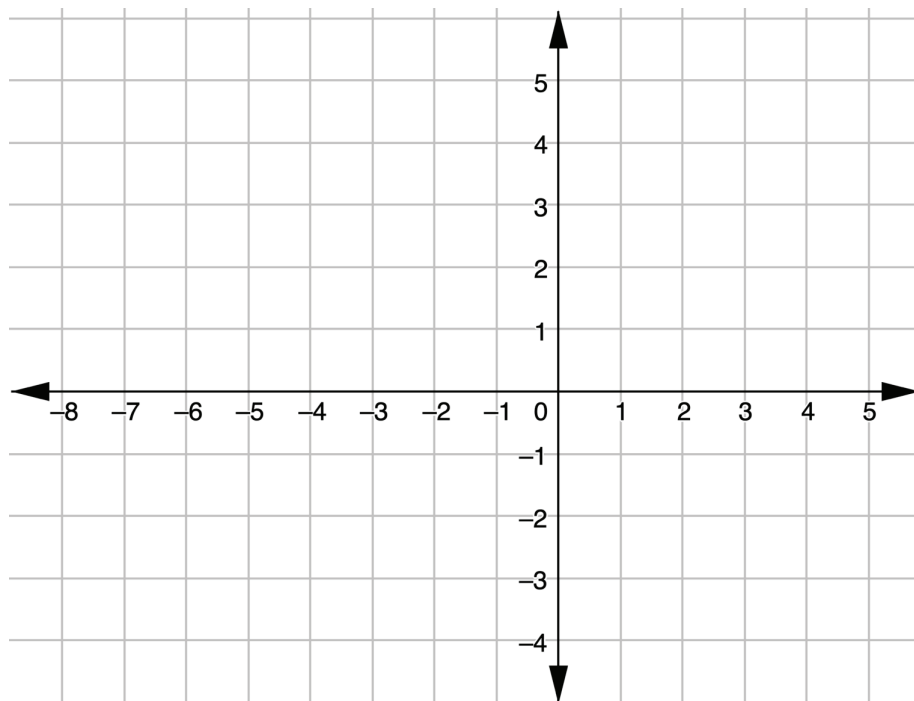
4. Draw a scalene triangle with one vertex at $(-3, 6)$ and with a base that is twice the length of its height.





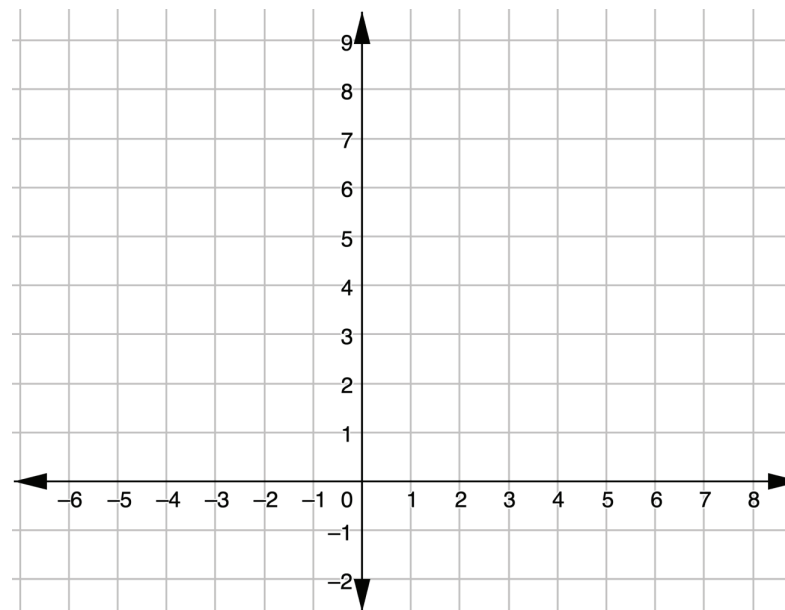
13.6.4 Your Turn!

1. $\triangle PTH$ has vertices at $P(-2,3)$, $T(3,1)$, and $H(0,-2)$. Determine if $\triangle PTH$ is an equilateral, isosceles, or scalene triangle. Justify your answer with math.



2. Given: $\triangle SCW$ with $W(1,6)$, $S(7,6)$, and $C(4,0)$

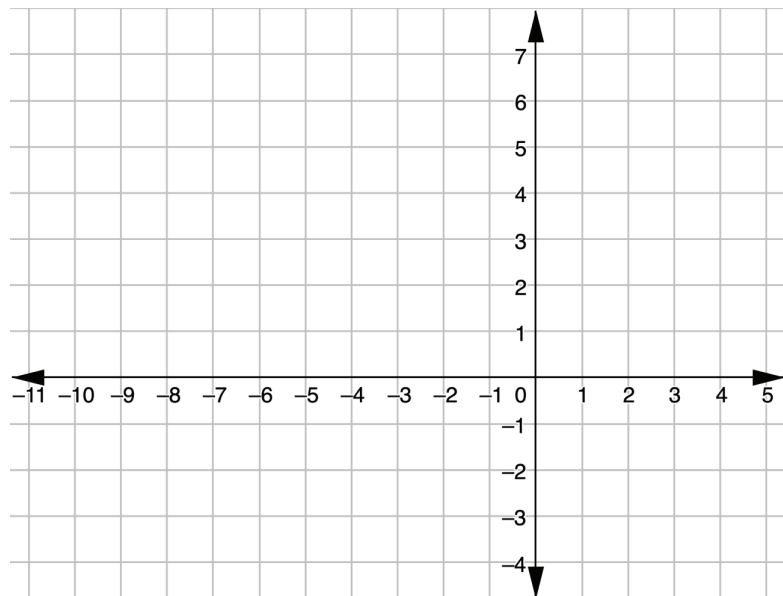
- a. Show: $\triangle SCW$ is **NOT** an equilateral triangle



- b. What type of triangle is $\triangle SCW$?

3. Diego is creating a triangular garden and uses the $\triangle YVX$ with vertices at $Y(-8, -2)$, $V(-6, 3)$, and $X(-2, -2)$ to design the garden.

a. Graph $\triangle YVX$.

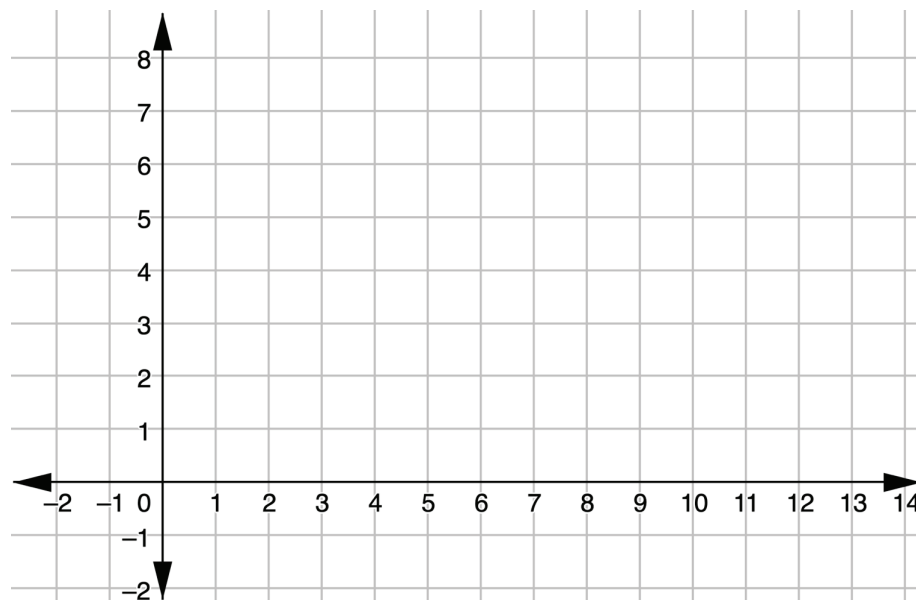


b. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Side Length
$\overline{YV}$	
$\overline{VX}$	
$\overline{XY}$	

c. Classify the type of triangular garden Diego is creating.

4. Draw an isosceles triangle whose base is three more than the height of the triangle and has one vertex at $(2, 5)$.





13.6.5 Wrap-Up: Cool Down

1. Identify **two** strategies you can use to classify triangles using coordinate geometry.



Unit 13, Lesson 7: Centroids of Triangles



Benchmark

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.



Learning Target

- ☐ I can use coordinate geometry to identify the centroid of a triangle.



13.7.1 Warm-Up: Human Hamster Wheel

A human hamster wheel is shown.

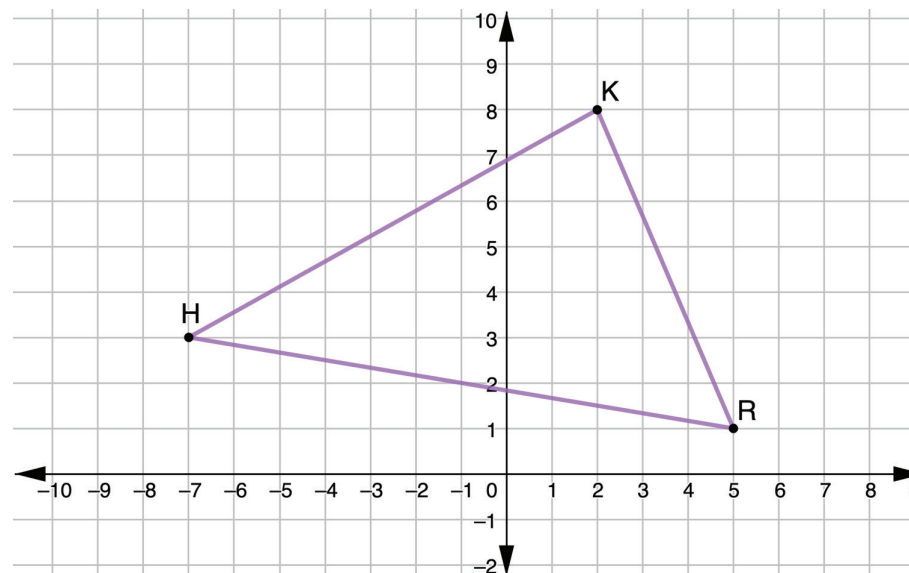


1. What is the first question that comes to your mind?
2. What are two things you notice about the human hamster wheel?



13.7.2 Exploration: Centroids of Triangles

1. $\triangle HKR$ has vertices at $H(-7,3)$, $K(2,8)$, and $R(5,1)$.



- a. Complete the table by finding the midpoint of each line segment.

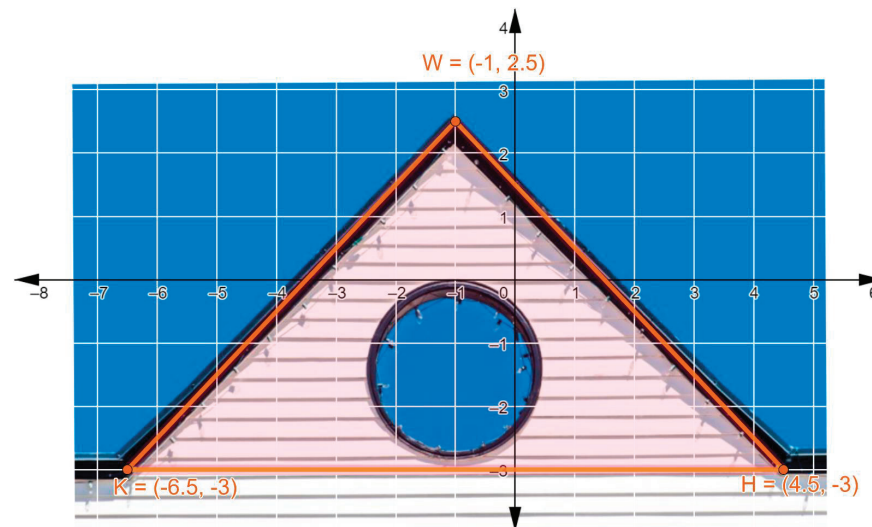
Line Segment	Work	Midpoint
$\overline{HK}$		
$\overline{KR}$		
$\overline{RH}$		

- b. Complete the steps to find the centroid of triangle HKR .

1	Graph the midpoint of line $\overline{HK}$ and label it W .
2	Using a straightedge, draw the median from W to the opposite vertex, $\angle HRK$.
3	Graph the midpoint of line $\overline{KR}$ and label it S .
4	Using a straightedge, draw the median from S to the opposite vertex, $\angle KHR$.
5	Graph the midpoint of line $\overline{RH}$ and label it M .
6	Using a straightedge, draw the median from M to the opposite vertex, $\angle HKR$.
7	Label the centroid of triangle HKR point P .

- c. What are the coordinates of the centroid of $\triangle HKR$?

2. The façade of a house is shown on the coordinate grid. The façade is represented by $\triangle WHK$ with vertices $W(-1, 2.5)$, $H(4.5, -3)$, and $K(-6.5, -3)$.



Work with your partner to estimate the location of the centroid of the triangular façade.

3. Use the centroids for $\triangle HKR$ and $\triangle WHK$ to complete the table.

Triangle	Centroid	Average of the x -coordinates of the Vertices	Average of the y -coordinates of the Vertices
$\triangle HKR$			
$\triangle WHK$			

4. Work with your partner to make a conjecture about the relationships of the centroid and the average of the x and y coordinates.

5. Ask two classmates for their conjectures. Record their conjectures and write your summary of their conjectures. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

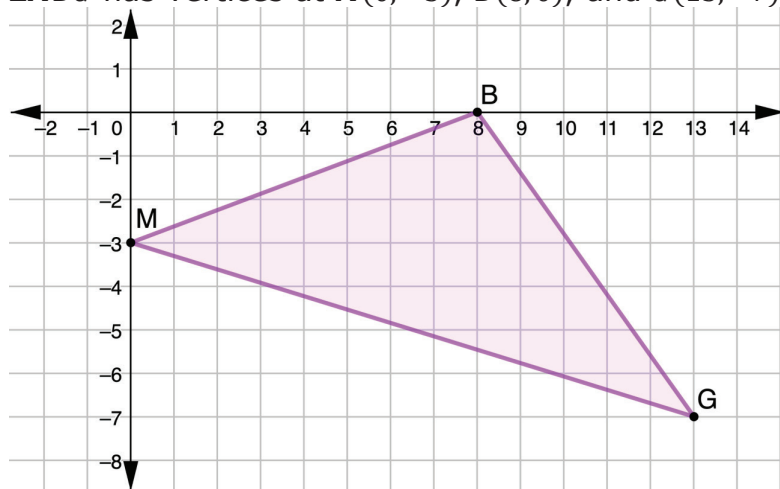
Conjecture	My Summary of Their Conjecture	Initials

6. Review your conjecture with your partner. If necessary, make any changes.



13.7.3 Collaboration: Centroids of Triangles

1. $\triangle MBG$ has vertices at $M(0, -3)$, $B(8, 0)$, and $G(13, -7)$.

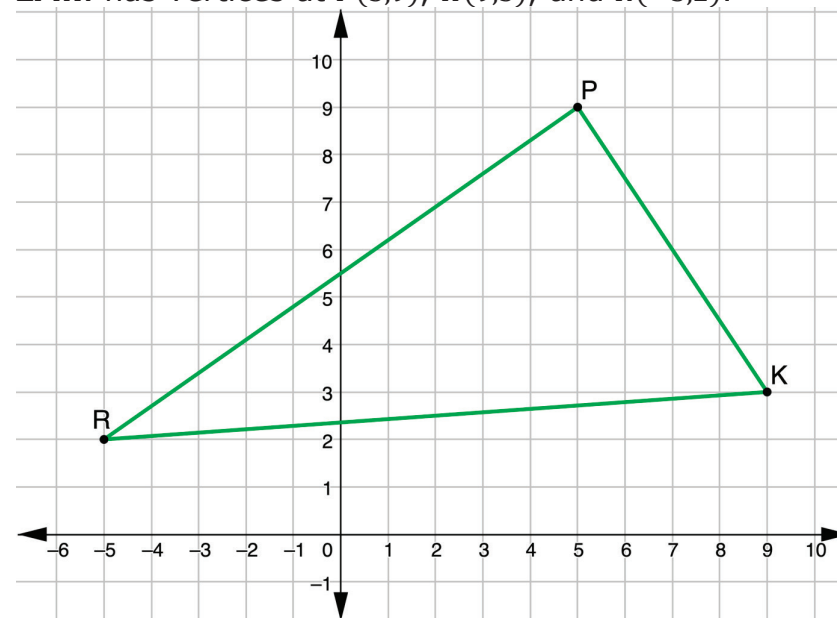


- What is the average of the x -coordinates?
- What is the average of the y -coordinates?
- What are the coordinates of the centroid?
- Verify the coordinates of the centroid by drawing the median of each side.



13.7.4 Your Turn!

1. $\triangle PKR$ has vertices at $P(5, 9)$, $K(9, 3)$, and $R(-5, 2)$.



What are the coordinates of the centroid of $\triangle PKR$?

2. $\triangle NHT$ has vertices at $N(-8, -3)$ and $H(3, -1)$ and a centroid at $(-2, -4)$.

What are the coordinates of vertex T ?

3. Artemio is finding the centroid of $\triangle WYV$ with vertices at $W(-8, 5)$, $Y(7, 8)$, and $V(-2, -3)$.

His work is shown.

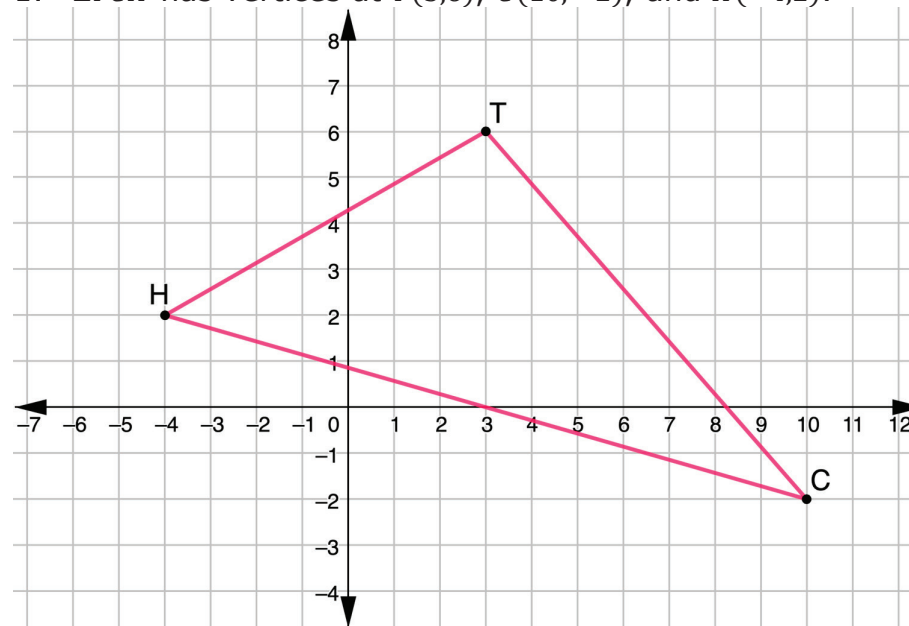
Artemio's Work	
$x = \frac{8 + 7 + 2}{3}$	$y = \frac{5 + 8 + 3}{3}$
$x = \frac{17}{3}$	$y = \frac{16}{3}$
Centroid $\left(\frac{17}{3}, \frac{16}{3}\right)$	

- Write a note to Artemio explaining his mistake.
- What are the coordinates of the centroid of $\triangle WYV$?



13.7.5 Wrap-Up: Ticket Out the Door

1. $\triangle TCH$ has vertices at $T(3, 6)$, $C(10, -2)$, and $H(-4, 2)$.



What are the coordinates of the centroid of $\triangle TCH$?

Unit 13, Lesson 8: Classifying Quadrilaterals – Part 1



Benchmark

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.



Learning Targets

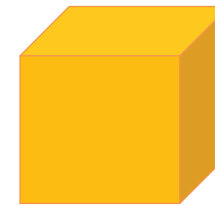
- ☐ I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.
- ☐ I can draw a quadrilateral on a coordinate plane using given definitions, properties, or theorems.



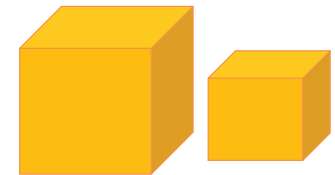
13.8.1 Warm-Up: Would You Rather?

Gold is worth a significant amount of money. Its value changes with the market. For example, on January 28, 2021 the value of 1 ounce of gold was nearly \$2000.

1. Would you rather have one cube made of solid gold measuring 14 meters (m) on each side, or two cubes made of solid gold, one measuring 12 m on each side and one measuring 8 m on each side?



14 m



12 m

8 m

2. Explain your choice.



13.8.2 Exploration: Classifying Quadrilaterals Using Properties

1. Complete the table by filling in the blanks using the word bank. Words may be used more than once, or they may not be used.

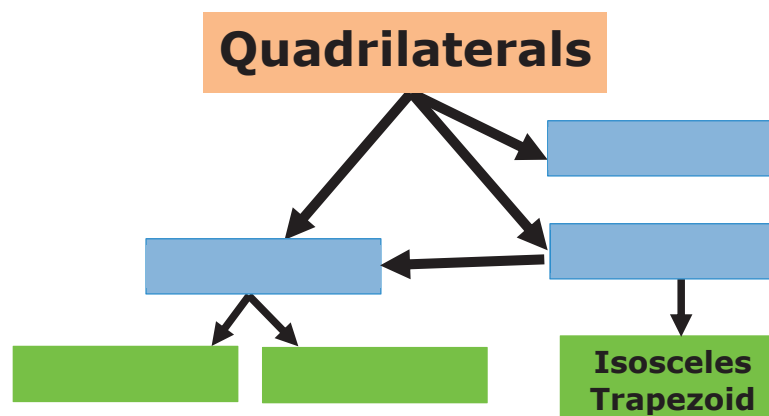
Word Bank	
bisect(s)	congruent
parallel	perpendicular

Classification	Property
Parallelogram	Opposite sides are _____ and _____.
Parallelogram	Diagonals _____ each other.
Rectangle	Diagonals are _____.
Rhombus	Diagonals are _____.
Square	Diagonals are _____ and _____.
Kite	Exactly one diagonal _____ the other, and diagonals are _____.
Isosceles Trapezoid	Diagonals are _____ and at least one pair of sides are _____.

2. Discuss with your partner if knowing two consecutive sides are congruent is sufficient to classify a quadrilateral. Summarize your discussion and justify your decision.

3. Discuss with your partner if knowing a pair of opposite sides are congruent is sufficient to classify the quadrilateral. Summarize your discussion and justify your decision.

4. Complete the diagram using the different types of quadrilaterals. Each blue box is a subcategory of quadrilaterals. Each green box is a subcategory of the blue box that it is connected to.



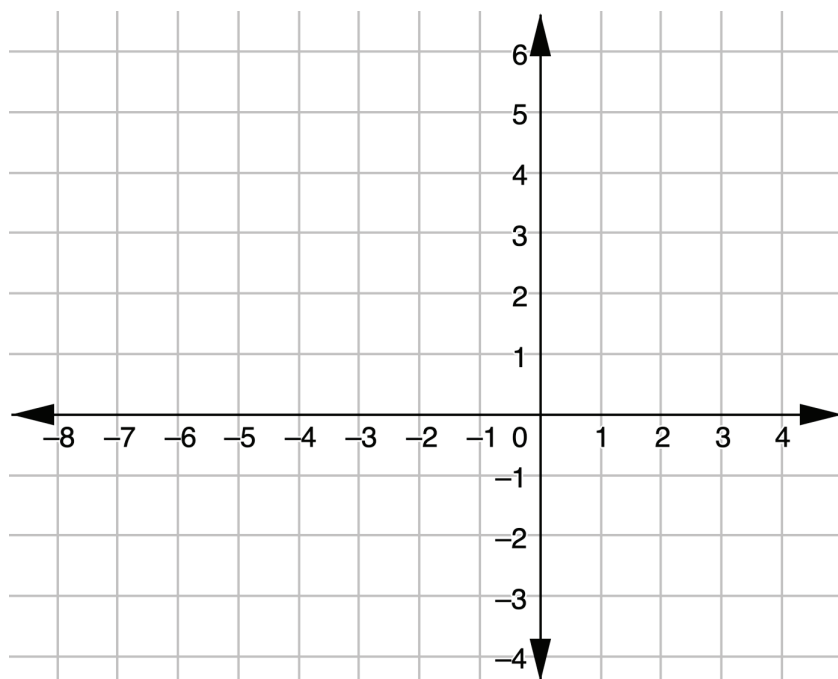
5. Which quadrilateral was not included in the diagram?
6. Draw a box and any necessary arrows to include the missing quadrilateral in the diagram.



13.8.3 Exploration: Classifying Quadrilaterals

1. $RMWB$ has vertices at $R(-6,2)$, $M(-4,5)$, $W(-1,3)$, and $B(-3,0)$.

- a. Graph $RMWB$ on the coordinate plane.



- b. Complete the table.

Line Segment	Length
$\overline{RM}$	
$\overline{MW}$	
$\overline{WB}$	
$\overline{BR}$	

- c. What do you notice about the side lengths of $RMWB$?

- d. Select all of the classifications that apply to $RMWB$.

- ☐ Isosceles trapezoid
- ☐ Kite
- ☐ Parallelogram
- ☐ Rectangle
- ☐ Rhombus
- ☐ Square
- ☐ Trapezoid

- e. Complete the table.

Diagonal Line Segment	Length
$\overline{RW}$	
$\overline{MB}$	

- f. What do you notice about the diagonal lengths of $RMWB$?

g. Complete the statement.

Since $RMWB$ ☐ has
☐ does not have congruent

diagonals, $RMWB$ is classified as a(n)

_____.

2. $NTCV$ has vertices at $N(2,3)$, $T(3,5)$, $C(6,6)$, and $V(4,2)$.

a. Complete the table.

Line Segment	Length
$\overline{NT}$	
$\overline{TC}$	
$\overline{CV}$	
$\overline{VN}$	

b. What do you notice about the side lengths of $NTCV$?

c. Complete the table.

Line Segment	Slope
$\overline{NT}$	
$\overline{TC}$	
$\overline{CV}$	
$\overline{VN}$	

d. What do you notice about the slopes of $NTCV$?

e. Select all of the classifications that apply to $NTCV$.

- ☐ Isosceles trapezoid
- ☐ Kite
- ☐ Parallelogram
- ☐ Rectangle
- ☐ Rhombus
- ☐ Square
- ☐ Trapezoid



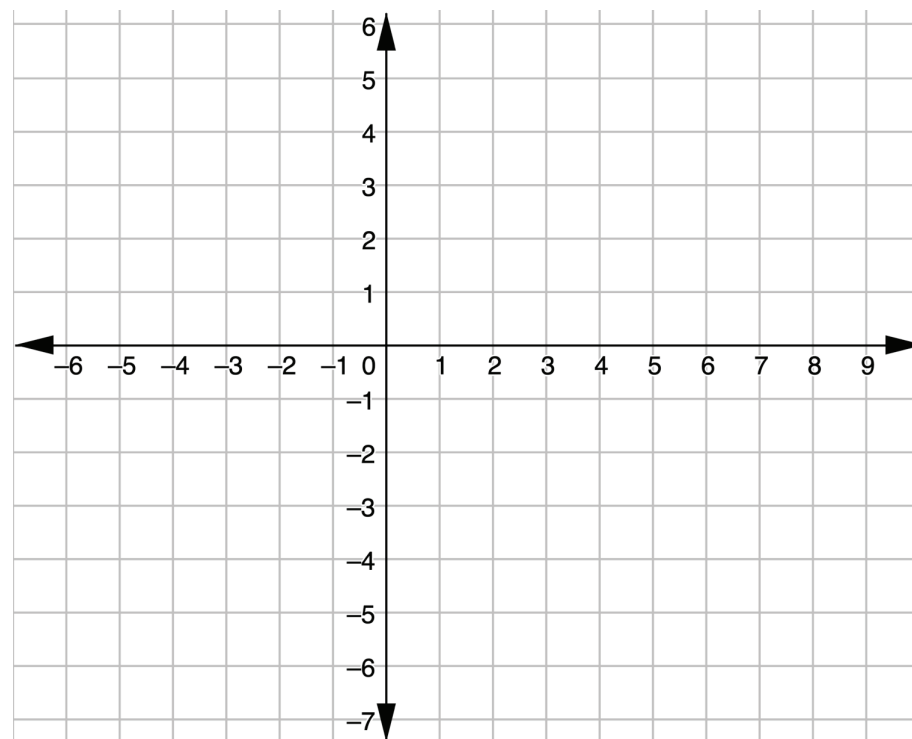
13.8.4 Guided Practice: Classifying Quadrilaterals

1. $GSPD$ has vertices at $G(-2, -3)$, $S(3, -2)$, $P(5, -4)$, and $D(0, -5)$.

Justify that $GSPD$ is a parallelogram but not a rectangle. Show your work.

2. $HMTK$ has vertices at $H(-2, 4)$, $M(-2, 0)$, $T(-6, -1)$, and $K(-6, 5)$. Classify $HMTK$ and justify your classification using mathematics.

3. Use the coordinate grid to complete the following.
- a. Draw a quadrilateral for which the side lengths are all equal, but the diagonals are not congruent.



- b. Complete the statement.

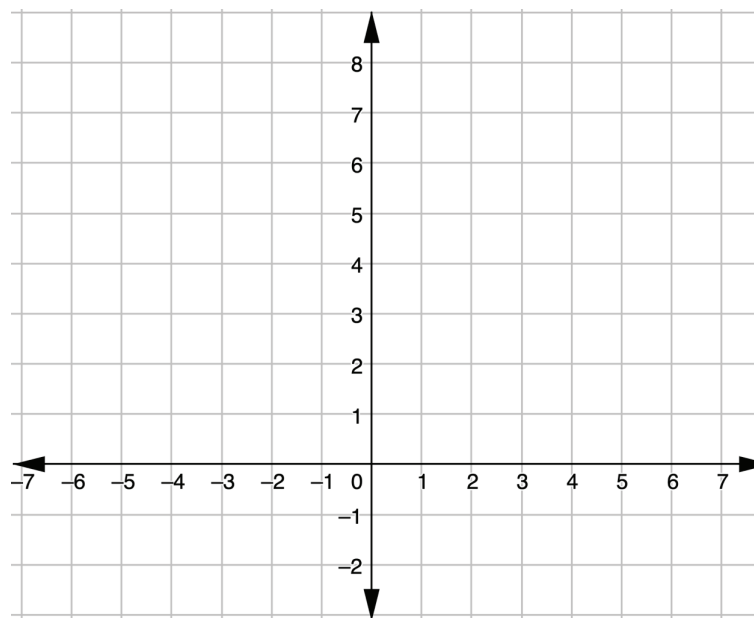
This quadrilateral is classified as a _____.

4. Discuss with your partner how another classmate could use the coordinates of your quadrilateral to verify that your classification is correct. Summarize your discussion.



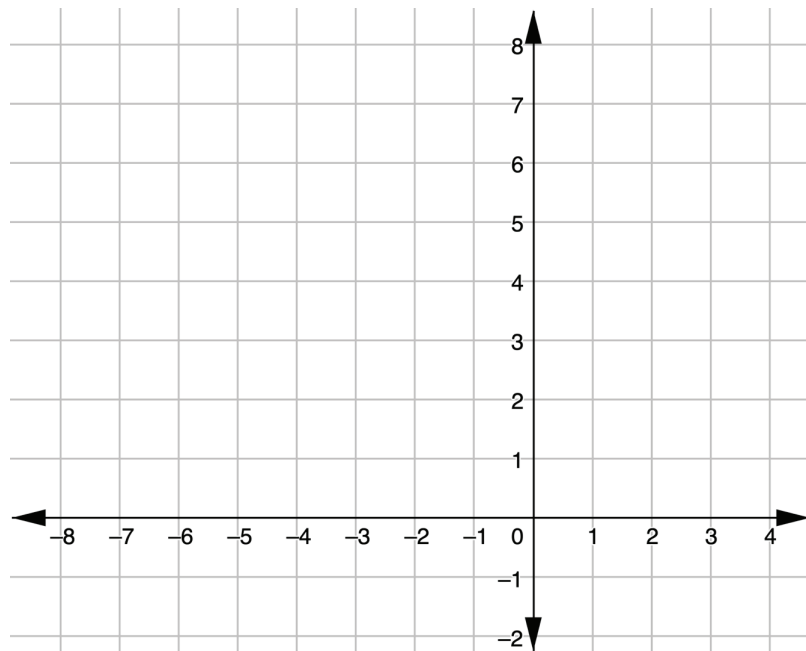
13.8.5 Your Turn!

1. $KHCY$ has vertices at $K(1,6)$, $H(3,4)$, $C(1,1)$, and $Y(-1,4)$.



$KHCY$ is classified as a _____ because ...

2. Draw a quadrilateral with one set of parallel sides and vertices at point $(-4,7)$ and $(-3,5)$.



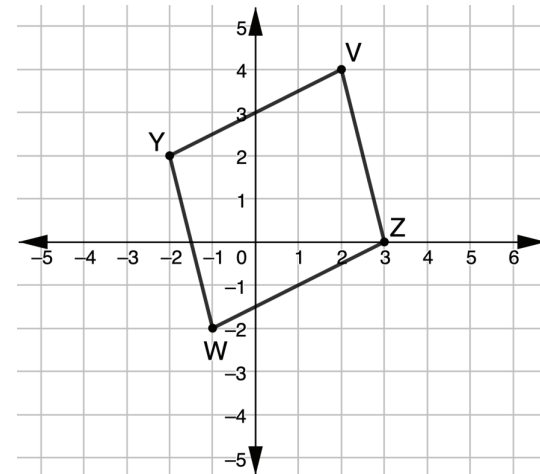
This quadrilateral is classified as a _____ because ...



13.8.6 Wrap-Up: Error Analysis

1. $YVZW$ has vertices at $Y(-2,2)$, $V(2,4)$, $Z(3,0)$, and $W(-1,-2)$.

Aryana graphed $YVZW$ and classified the quadrilateral as a rhombus.



- a. Do you agree with Aryana?
- b. Justify your answer using properties and showing your work.

Unit 13, Lesson 9: Classifying Quadrilaterals – Part 2



Benchmarks

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.



Learning Targets

- ☐ I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.
- ☐ I can use coordinate geometry to solve geometric problems involving quadrilaterals in a mathematical or real-world context.



13.9.1 Warm-Up: Between Two Numbers

Sigsbee Deep	Britton Hill	Panorama Tower
Dwight Sigsbee led the first expedition to chart the floor of the Gulf of Mexico. The Sigsbee Deep, the deepest region of the Gulf of Mexico, was named after him. The exact maximum depth is not known, but is estimated to be between 3,750 meters (m) and 4,384 m.	Britton Hill, located in Walton County, is the highest point in the state of Florida. Britton Hill's elevation is 105 m.	Panorama Tower in Miami is the tallest building in Florida. It is 265 m tall.

- Complete the inequality using the digits 0 – 9, without repeating, where B = number of Britton Hills and P = number of Panorama Towers.

$$3750 < \boxed{}\boxed{}\boxed{}\boxed{}B + \boxed{}\boxed{}\boxed{}\boxed{}P < 4384$$

- Use a different set of digits, without repeating.

$$3750 < \boxed{}\boxed{}\boxed{}\boxed{}B + \boxed{}\boxed{}\boxed{}\boxed{}P < 4384$$



13.9.2 Collaboration: Real-World Quadrilaterals

1. Francesco is using computer software to design the layout of a house. The base of the house creates $DMHT$ with coordinates $D(-4, 1)$, $M(4, 5)$, $H(6, 1)$, and $T(-2, -3)$.

- a. Complete the table. Show your work in the table. Leave answers in exact form.

Line Segment	Length
$\overline{DM}$	
$\overline{MH}$	
$\overline{HT}$	
$\overline{TD}$	

- b. Complete the table. Show your work in the table. Leave answers in exact form.

Diagonal Line Segment	Length
$\overline{DH}$	
$\overline{MT}$	

- c. Finish the statement.

$DMHT$ is classified as a _____ because

- d. If 1 unit on the coordinate plane equals 5 feet (ft), what are the dimensions of the base of the house? Round to the nearest thousandth.

2. Jillian is determining which type of tile to place on the floor. She wants a tile in the shape of a parallelogram. The vertices of the tile options are shown.

Tile A	Tile B
$(-11,6.5)$, $(-10,3.5)$, $(-6,6.5)$, and $(-7,9.5)$	$(3,9.5)$, $(7,6)$, $(9,8.5)$, and $(5,11)$

- a. Which tile should Jillian pick if she wants a tile in the shape of a parallelogram?
- b. Explain your choice by listing the properties of parallelograms that were applied.

3. Aharon and Shreya classified $RNPF$ with vertices at $R(-3,7)$, $N(-1,7)$, $P(-3,2)$, and $F(-5,4)$. Their classifications are shown.

Aharon's Classification	Shreya's Classification
$RNPF$ is a square or rectangle because the diagonals are congruent.	Quadrilateral $RNPF$ has congruent diagonals but cannot be classified as a parallelogram.

- a. Whose classification do you agree with? Justify your answer using properties.
- b. Write a note to the person whose classification is incorrect, explaining their mistake.

4. $KVSW$ has vertices at $K(10, -4)$, $V(9, -8)$, $S(1, -4)$, and $W(6, -2)$.
- a. Show mathematically that $KVSW$ is a trapezoid, but not an isosceles trapezoid.

- b. List the properties of an isosceles trapezoid that did not apply to $KVSW$.

5. Ariyon was graphing square $MGNR$, but mixed up her vertices. She knows that point M has coordinates $(3, 5)$ and point G has coordinates $(7, 7)$. Her two options for points N and R are shown.

Point N	Point R
$(8, 3)$	$(5, 1)$
$(9, 3)$	$(4, 1)$

- a. Which coordinates are correct for points N and R ? Justify your answer or show your work.
- b. Which properties of a square did you use to determine the correct points N and R ?



13.9.3 Your Turn!

1. $SCWK$ has vertices at $S(-3, -4)$, $C(0, -2)$, $W(0, -10)$, and $K(-3, -8)$.

$SCWK$ is classified as a(n) _____
because ...

2. Ziquan is designing a template for his 3D printer. He created two quadrilaterals using his software.

<i>TGHN</i>	<i>MWRB</i>
$T(16, 6)$, $G(18, 4)$, $H(17, -2)$, and $N(14, 4)$	$M(4, 5)$, $W(7, 4)$, $R(8, -3)$, and $B(3, 2)$

- a. Which quadrilateral can be classified as a kite?
Show your work.

- b. List the properties of a kite that you used.

- c. If 1 unit on the coordinate plane equals 6 inches (in), what are the measurements of the sides of the kite? Round to the nearest thousandth.



13.9.4 Wrap-Up: Reflection

1. Place an **X** on the line in relation to how you feel about today's learning target.

- ☐ I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.

I need help.
I can't get started.

I'm getting there.

I understand and have
evidence.



Unit 13, Lesson 10: Classifying Quadrilaterals – Part 3



Benchmark

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.



Learning Target

- ☐ I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.



13.10.1 Warm-Up: Would You Rather?

1. Would you rather ...

Give up watching
movies?

OR

Give up watching
TV shows?

2. Explain your choice.



13.10.2 Activity: Stations

Complete the table for each question card.

Question	Work		Answer
			Properties
1	Slopes		
	Side Lengths		
	Diagonal Lengths		Quadrilateral

Question	Work		Answer
			Properties
2	Slopes		
	Side Lengths		
	Diagonal Lengths		Quadrilateral

Question	Work		Answer
3	Length of $\overline{WH}$ and $\overline{HC}$		
	Slopes of $\overline{WH}$ and $\overline{HC}$		Verification
	Slope of $\overline{CS}$ Using Each Point S		

Question	Work		Answer
			Properties
4	Slopes		
	Side Lengths		
	Diagonal Lengths		Who is correct?

Question	Work		Answer
			Properties
5	Slopes		
	Side Lengths		
	Diagonal Lengths		Quadrilateral

Question	Work		Answer
6	Slopes		Which quadrilateral is a rectangle?
	Side Lengths		Properties
	Diagonal Lengths		

Question	Work		Answer
7	Slopes		Why is $VWXY$ not a square?
	Side Lengths		
	Diagonal Lengths		



13.10.3 Wrap-Up: Cool Down

- 1. Write a short note to a friend who missed today’s class summarizing what you learned.

Question	Work		Answer
8	Slope and Length of $\overline{KD}$		
	Slope of $\overline{MS}$ Using Each Point S		

Verification



Unit 13, Lesson 11: What is the Figure?



Benchmarks

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.



Learning Targets

- ☐ I can use coordinate geometry to classify triangles and quadrilaterals.
- ☐ I can use coordinate geometry to justify definitions, properties, or theorems of triangles and quadrilaterals.
- ☐ I can use coordinate geometry to solve mathematical problems involving triangles and quadrilaterals.



13.11.1 Warm-Up: Bell Work

Jonah is designing a stage in the shape of a trapezoid. Jonah has created two designs.

<i>VSZY</i>	<i>BGKR</i>
$V(2, -5)$, $S(4, -7)$, $Z(-2, -7)$, and $Y(-1, -5)$	$B(-4, -2)$, $G(-2, -2)$, $K(-7, -7)$, and $R(-7, -5)$

1. Determine whether Jonah can use either or both designs for the trapezoidal stage.
 - Show your work in an organized way.
 - Write your conclusion explaining the properties used to determine your answer.



13.11.2 Activity: Card Sort

Match each image, classification, coordinates, and property that represent the same polygon. Record your matches in the table by writing the letter or number of each card and showing your justifications.

Match 1	Match 2
Image:	Image:
Coordinates:	Coordinates:
Classification:	Classification:
Property:	Property:
Justification:	Justification:

Match 3	Match 4
Image:	Image:
Coordinates:	Coordinates:
Classification:	Classification:
Property:	Property:
Justification:	Justification:

Match 5	Match 6
Image:	Image:
Coordinates:	Coordinates:
Classification:	Classification:
Property:	Property:
Justification:	Justification:

Match 7	Match 8
Image:	Image:
Coordinates:	Coordinates:
Classification:	Classification:
Property:	Property:
Justification:	Justification:

Match 9	Match 10
Image:	Image:
Coordinates:	Coordinates:
Classification:	Classification:
Property:	Property:
Justification:	Justification:



13.11.3 Wrap-Up: Cool Down

1. Explain how you can use coordinate geometry to classify triangles and quadrilaterals.

Unit 13, Lesson 12: Congruence and Similarity



Benchmark

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.



Learning Target

- ☐ I can use coordinate geometry to justify congruence and similarity in triangles and quadrilaterals.



13.12.1 Warm-Up: Notice and Wonder

Brickley, the LEGO sea serpent in Lake Buena Vista at Disney Springs near Orlando, Florida, was made in 1997 from 170,000 LEGO bricks and is 30 feet long.



1. What do you notice?
2. What do you wonder?



13.12.2 Exploration: Congruence and Similarity

1. Parallelogram $DKTS$ has vertices at $D(-2,4)$, $K(-4,1)$, $T(-9,3)$, and $S(-7,6)$.

Complete the table.

Line Segment	Length
$\overline{DK}$	
$\overline{KT}$	
$\overline{TS}$	
$\overline{SD}$	

2. Parallelogram $WRCN$ has vertices at $W(0,1)$, $R(2,-2)$, $C(7,0)$, and $N(5,3)$.

Complete the table.

Line Segment	Length
$\overline{WR}$	
$\overline{RC}$	
$\overline{CN}$	
$\overline{NW}$	

3. What do you notice about the lengths of the line segments in parallelograms $DKTS$ and $WRCN$?

4. Complete the statement.

Parallelogram $DKTS$ may be

- ☐ congruent
- ☐ similar

to parallelogram $WRCN$ because all corresponding sides

are

- ☐ proportional.
- ☐ congruent.

5. Work with your partner to describe how the parallelograms are congruent or similar using transformations. State which image you used as the preimage.



6. Isosceles trapezoid $MPHN$ has vertices at $M(2, -6)$, $P(3, -3)$, $H(5, -2)$, and $N(8, -3)$.

Complete the table.

Line Segment	Length
$\overline{MP}$	
$\overline{PH}$	
$\overline{HN}$	
$\overline{NM}$	

7. Isosceles trapezoid $WVYZ$ has vertices at $W(4, 12)$, $V(6, 6)$, $Y(10, 4)$, and $Z(16, 6)$.

Complete the table.

Line Segment	Length
$\overline{WV}$	
$\overline{VY}$	
$\overline{YZ}$	
$\overline{ZW}$	

8. What do you notice about the lengths of the line segments in isosceles trapezoids $MPHN$ and $WVYZ$?

9. Complete the statement.

Isosceles trapezoid $MPHN$ is
☐ congruent
☐ similar
 to

isosceles trapezoid $WVYZ$ because all corresponding

sides are
☐ proportional.
☐ congruent.

10. Work with your partner to describe how the trapezoids are congruent or similar using transformations. State which image you used as the preimage.



13.12.3 Guided Practice: Congruence and Similarity

1. Right triangle CKR has vertices at $C(-2,5)$, $K(-8,5)$, and $R(-8,2)$. Right triangle DPM has vertices at $D(-2,-4)$, $P(4,-4)$, and $M(4,-1)$.
 - a. Determine if $\triangle CKR$ and $\triangle DPM$ are congruent, similar, or neither.
 - b. What definition, theorem, or property justifies your conclusions?
 - c. Describe the transformations that would show that the figures are congruent or that they are similar. State which image you used as the preimage.

- d. Ask a classmate for their description. Record their description and write your summary of their description. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Description	My Summary of Their Description	Initials

2. Parallelogram $HGNC$ has vertices at $H(5, -1)$, $G(2, 3)$, $N(9, 3)$, and $C(12, -1)$. Parallelogram $VYZW$ has vertices at $V(-1, 1.5)$, $Y(0.5, 4)$, $Z(-3, 4)$, and $W(-4.5, 1.5)$.

- a. Shin and Kendal are determining if parallelograms $HGNC$ and $VYZW$ are congruent, similar, or neither. Their work is shown.

Shin's Work	Kendal's Work
Parallelograms $HGNC$ and $VYZW$ are similar because $\frac{ZY}{NG} = \frac{VW}{HC}$	Parallelograms $HGNC$ and $VYZW$ are neither because $\frac{ZY}{NG} \neq \frac{YV}{GH}$

- b. Whose work do you agree with?
- c. Write a note to the person who is incorrect explaining their mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills



13.12.4 Your Turn!

Complete the table by determining if the given polygons are congruent, similar, or neither.

1. $\triangle STK$: $S(-6,8)$, $T(-10,5)$, and $K(-6,1)$

$\triangle NRC$: $N(3,7.5)$, $R(-3,3)$, and $C(3,-3)$

- ☐ Congruent
- ☐ Similar
- ☐ Neither

2. Trapezoid $MPWY$: $M(3,4)$, $P(7,6)$, $W(10,5)$, and $Y(4,2)$

Trapezoid $LKGH$: $L(-2,6)$, $K(-8,9)$, $G(-14,7)$, and $H(-4,2)$

- ☐ Congruent
- ☐ Similar
- ☐ Neither

3. Kite $XZWY$: $X(-7,6)$, $Z(-6,9)$, $W(-3,10)$, and $Y(-1,4)$

Kite $FGDK$: $F(2,3)$, $G(1,6)$, $D(-2,7)$, and $K(-4,1)$

- ☐ Congruent
- ☐ Similar
- ☐ Neither



13.12.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to teach them how to use coordinate geometry to justify congruence and similarity in polygons.

Unit 13, Lesson 13: Finding Perimeter and Area on the Coordinate Plane



Benchmark

MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving the perimeter or area of polygons.



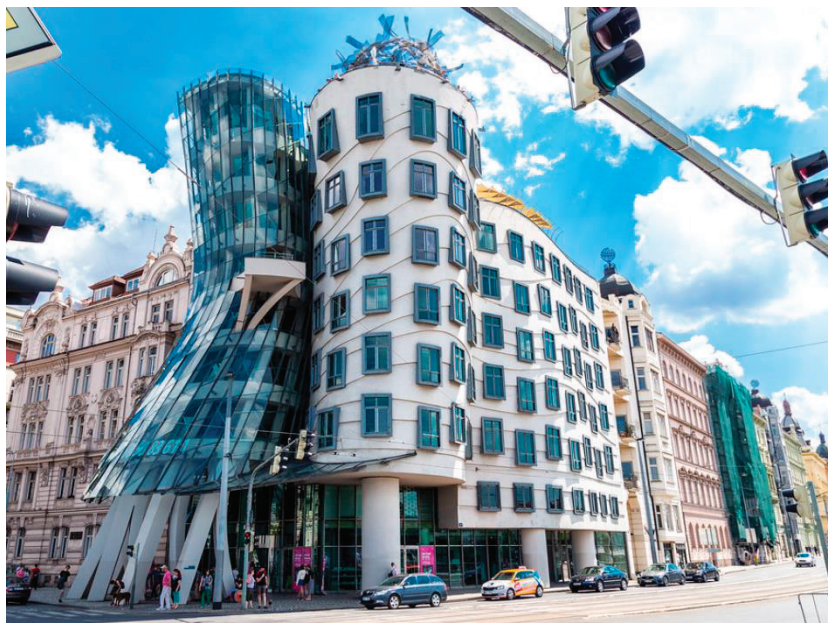
Learning Target

- ☐ I can find the perimeter and area of polygons and composite figures drawn on the coordinate plane.



13.13.1 Warm-Up: The Dancing House

The Dancing House is located in Prague, Czech Republic, and was designed in 1992. The style of this building is known as deconstructivist architecture due to its unusual shape.

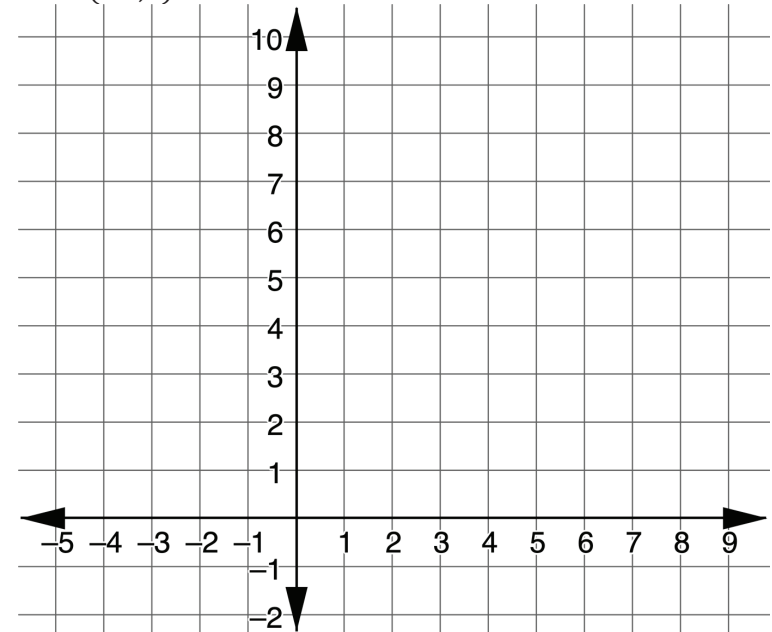


1. What is the first question that comes to your mind?
2. Write a question that a 5th grader could answer using the photo.



13.13.2 Exploration: Perimeter and Area

1. Trapezoid $BGMH$ has vertices at $B(1,9)$, $G(5,9)$, $M(2,5)$, and $H(-4,5)$.



- a. Graph $BGMH$.
- b. Complete the table.

Line Segment	Length
$\overline{BG}$	
$\overline{GM}$	
$\overline{MH}$	
$\overline{HB}$	

c. What is the perimeter of $BGMH$?

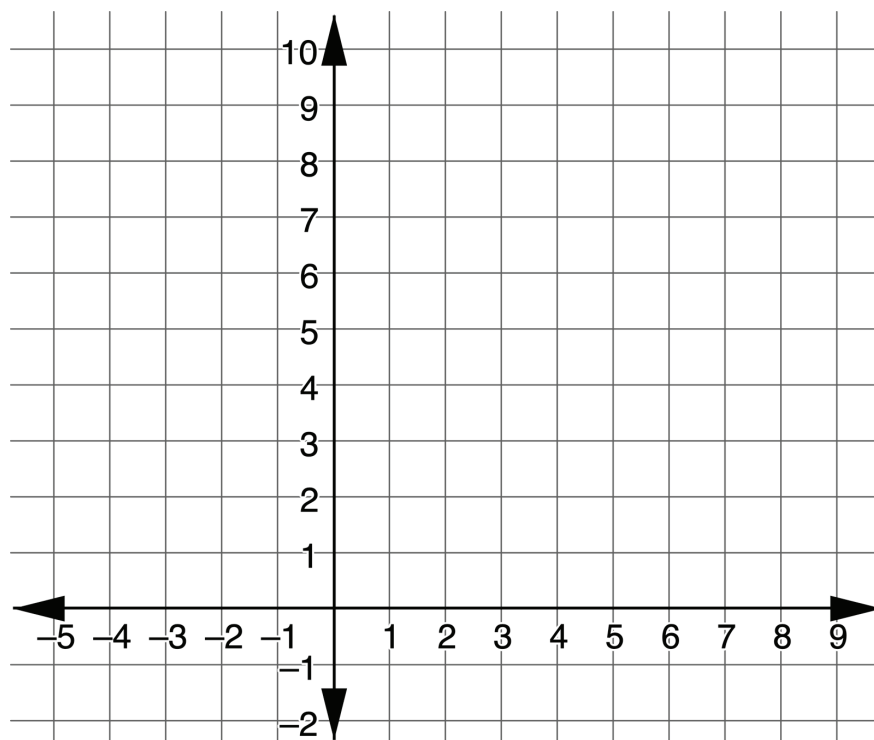
d. Draw a diagonal line from B to M to make $\triangle BMH$ and $\triangle BGM$.

e. Find the area of $\triangle BMH$ and $\triangle BGM$.

$\triangle BMH =$ _____ $\triangle BGM =$ _____

f. What is the area of $BGMH$?

2. Parallelogram $GPKM$ has vertices at $G(5,9)$, $P(5,6)$, $K(2,2)$, and $M(2,5)$.



a. Graph $GPKM$.

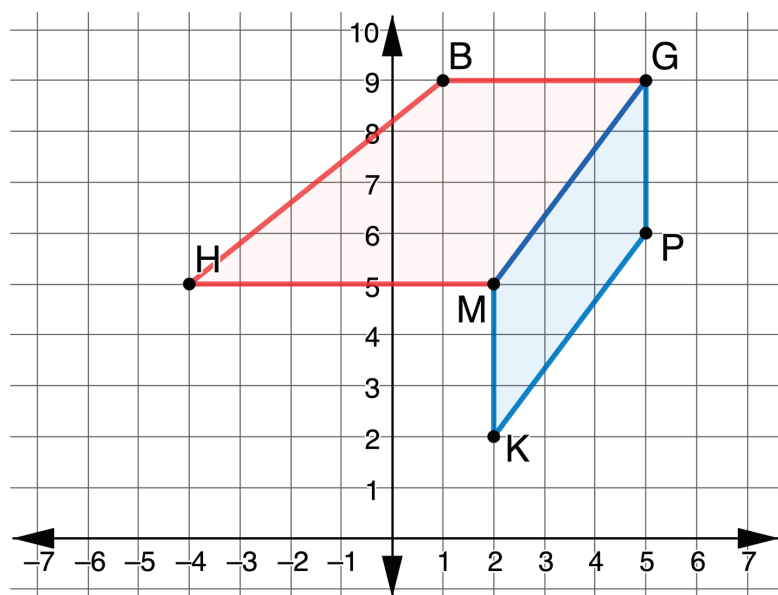
b. Complete the table.

Line Segment	Length
$\overline{GP}$	
$\overline{PK}$	
$\overline{KM}$	
$\overline{MG}$	

c. What is the perimeter of $GPKM$?

d. What is the area of $GPKM$?

3. The figure shown is created by trapezoid $BGMH$ and parallelogram $GPKM$.



a. What do you notice about this figure?



A **composite figure** is a two- or three-dimensional figure that can be decomposed into smaller figures.

b. Explain why this figure is considered a composite figure.

c. How could you find the perimeter of a composite figure?

d. What is the perimeter of composite figure $BGPCKMH$?

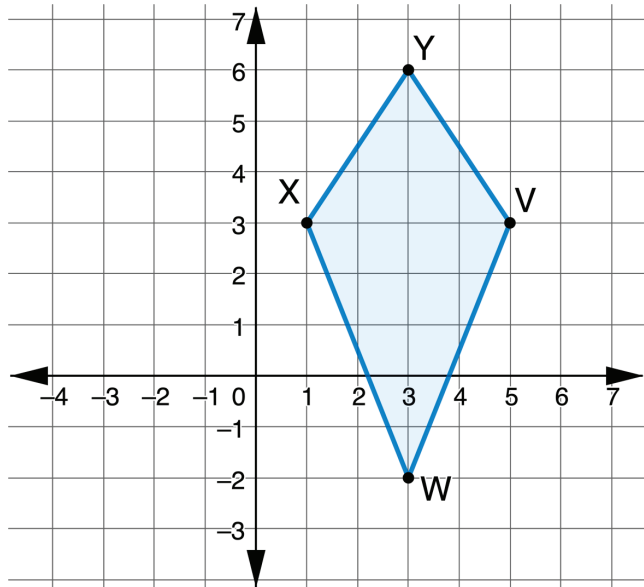
e. Explain why the perimeter of the composite figure is not the sum of the perimeters of the parallelogram and the trapezoid.

f. What is the area of $BGPCKMH$?



13.13.3 Try It: Perimeter and Area

1. Kite $YVWX$ is shown with vertices at $Y(3,6)$, $V(5,3)$, $W(3,-2)$, and $X(1,3)$.



- a. Complete the table.

Line Segment	Length
$\overline{YV}$	
$\overline{VW}$	
$\overline{WX}$	
$\overline{XY}$	

- b. What is the perimeter of $YVWX$?

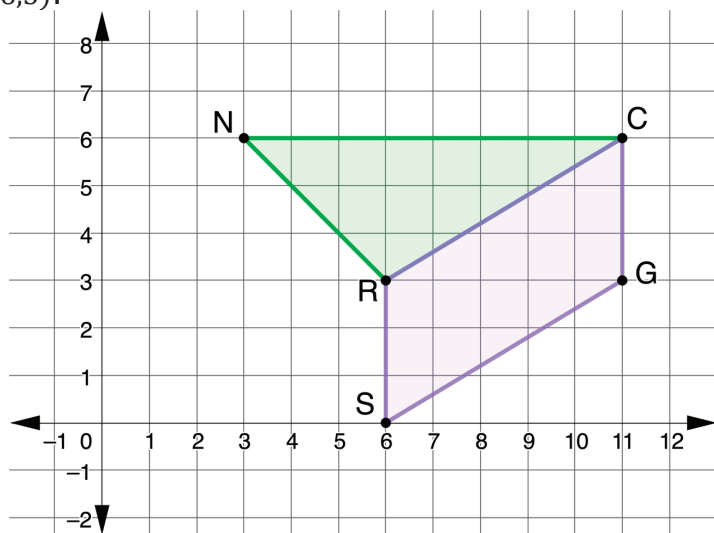
- c. Draw line XV to create $\triangle YVX$ and $\triangle VWX$.

- d. Find the area of each triangle.

$\triangle YVX =$ _____ $\triangle VWX =$ _____

- e. What is the area of $YVWX$?

2. The composite figure shown is created by $\triangle NCR$ and parallelogram $RCGS$. $N(3,6)$, $C(11,6)$, $G(11,3)$, $S(6,0)$, and $R(6,3)$.



- a. Complete the table. Round your answer to the nearest thousandth.

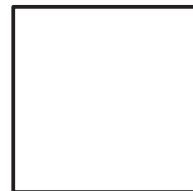
Figure	Perimeter	Area
$\triangle NCR$		
$RCGS$		

- b. What is the perimeter of the composite figure?
- c. What is the area of the composite figure?

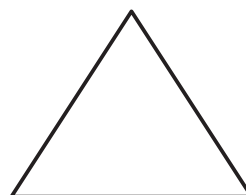


13.13.4 Wrap-Up: Cool Down

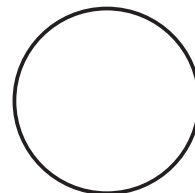
1. What is something that is now “squared away” in your mind about finding the perimeter and area of polygons and composite figures drawn on a coordinate plane?



2. What are three “points” that are important about finding the perimeter and area of polygons and composite figures drawn on a coordinate plane?



3. What is something that is still “circling” your brain about finding the perimeter and area of polygons and composite figures drawn on a coordinate plane?



Unit 13, Lesson 14: Using Coordinate Geometry to Solve Real-World Problems



Benchmarks

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.

MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons.



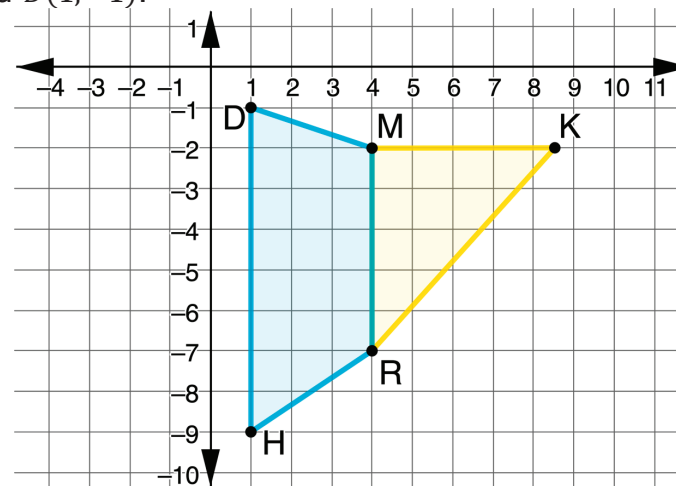
Learning Targets

- ☐ I can use coordinate geometry to classify triangles and quadrilaterals.
- ☐ I can use coordinate geometry to solve real-world problems on the coordinate plane involving perimeter or area of polygons.
- ☐ I can use coordinate geometry to solve real-world geometric problems involving circles, triangles, and quadrilaterals.



13.14.1 Warm-Up: Bell Work

1. The composite figure shown is created by $\triangle MKR$ and trapezoid $DMRH$. $M(4, -2)$, $K(8.5, -2)$, $R(4, -7)$, $H(1, -9)$, and $D(1, -1)$.



- a. Complete the table. If necessary, round to the nearest thousandth.

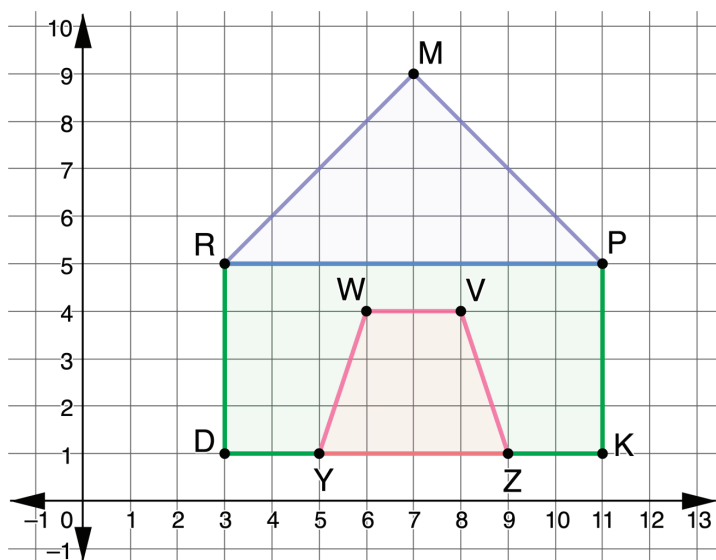
Figure	Perimeter	Area
$\triangle MKR$		
$DMRH$		

- b. What is the perimeter of the composite figure? If necessary, round to the nearest thousandth.
- c. What is the area of the composite figure? If necessary, round to the nearest thousandth.



13.14.2 Collaboration: Real-World Problems

1. Lana is designing the front of a doghouse. The computer software generated coordinate points where each unit equals a foot. The roof has vertices at $M(7,9)$, $P(11,5)$, and $R(3,5)$. The front wall has vertices at $P(11,5)$, $K(11,1)$, $D(3,1)$, and $R(3,5)$. The door entrance has vertices at $Z(9,1)$, $Y(5,1)$, $W(6,4)$, and $V(8,4)$.



- a. What type of triangle is the roof, $\triangle MPR$? Justify your answer.
- b. What is the area of $\triangle MPR$?

- g. Lana is painting the front wall, $RPKD$, excluding the door, $WVZY$. What is the area of the front wall Lana is painting?

- h. Paint costs \$0.55 per square foot. Excluding the door, how much will it cost to paint the front wall?

- g. Lana is painting the front wall, $RPKD$, excluding the door, $WVZY$. What is the area of the front wall Lana is painting?

- h. Paint costs \$0.55 per square foot. Excluding the door, how much will it cost to paint the front wall?

2. Justin is creating a layout for his garden in the shape of a quadrilateral, $GWDP$, with vertices $G(1,8)$, $W(11,8)$, $D(11,0)$, and $P(1,0)$.

- a. Classify $GWDP$. Justify your answer.

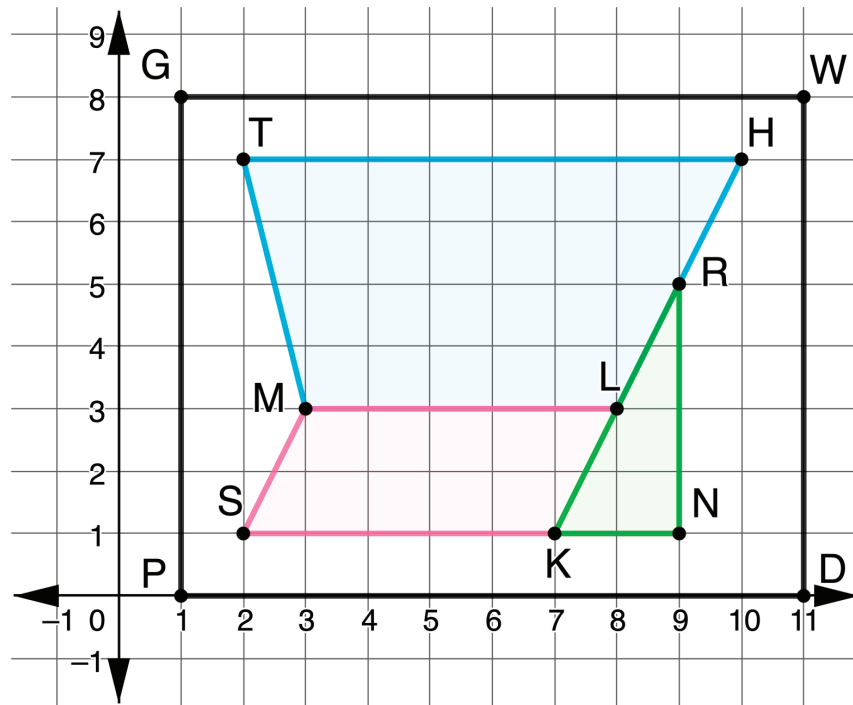
Justin created three sections in the garden: $THLM$, $LKSM$, and $\triangle KNR$.

- b. $THLM$ has vertices at $T(2,7)$, $H(10,7)$, $L(8,3)$, and $M(3,3)$. Classify $THLM$. Justify your answer.

- c. $LKSM$ has vertices at $L(8,3)$, $K(7,1)$, $S(2,1)$, and $M(3,3)$. Classify $LKSM$. Justify your answer.

- d. $\triangle KNR$ has vertices at $K(7,1)$, $N(9,1)$, and $R(9,5)$. Classify $\triangle KNR$. Justify your answer.

The layout of the garden is shown.



- e. If one square on the grid represents one square foot, how many feet of fencing will Justin need to go around $GWDP$?

- f. Complete the table. If necessary, round to the nearest thousandth.

Section	Perimeter	Area
$THLM$		
$LKSM$		
$\triangle KNR$		

- g. What is the perimeter of the composite figure made by the three sections of the garden? If necessary, round to the nearest thousandth.
- h. What is the area of the three sections of the garden? If necessary, round to the nearest thousandth.
- i. If a 40 pound (lb) bag of soil covers 12 square feet, how many bags of soil will Justin need to cover the three sections of the garden?

- a. Complete the table by finding the area of each triangle.

Triangle	Work	Area
$\triangle AHR$		
$\triangle AMG$		
$\triangle PGM$		
$\triangle TKM$		

- b. Explain how to use $\triangle PGM$ and $\triangle TKM$ to find the area of $TKPG$.
- c. What is the area of $TKPG$? If necessary, round to the nearest thousandth.
- d. What is the total area of Edwin's design? If necessary, round to the nearest thousandth.

- a. Complete the table by finding the area of each triangle.

Triangle	Work	Area
$\triangle AHR$		
$\triangle AMG$		
$\triangle PGM$		
$\triangle TKM$		

- b. Explain how to use $\triangle PGM$ and $\triangle TKM$ to find the area of $TKPG$.
- c. What is the area of $TKPG$? If necessary, round to the nearest thousandth.
- d. What is the total area of Edwin's design? If necessary, round to the nearest thousandth.



13.14.3 Wrap-Up: Reflection

Circle the icon that best describes your understanding of each.



I need help.



I got it.



I could help someone.

Using coordinate geometry to determine if lines are parallel or perpendicular.	  
Using coordinate geometry to partition a line segment or find the weighted average.	  
Using coordinate geometry to classify triangles or quadrilaterals.	  
Using coordinate geometry to justify definitions, properties, and theorems involving circles, triangles or quadrilaterals.	  
Using coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.	  
Using coordinate geometry to solve mathematical and real-world involving perimeter or area of polygons.	  

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